

DATA PIPELINE 06: MYSQL(ON-PREM) Data Migration-ADLS-ADF(DATAFLOW)-AZURE SQL

Project Objective:

The objective of this project is to design and implement a scalable ETL data pipeline that extracts data from multiple on-premises MySQL database tables, stages the data in Azure Data Lake Storage Gen2 (ADLS), performs data cleansing and business transformations using Azure Data Factory (ADF) Mapping Data Flows, and loads the transformed data into Azure SQL Database for reporting and analytics.

Process:

MySQL (On-Premises) → ADLS (Raw Layer) → ADF Mapping Data Flow (Transformation Layer) → Azure SQL Database

Pipeline 1 – Data Ingestion (On-Premises → ADLS)

Purpose: Copy raw data from the on-premises MySQL database to Azure Data Lake Storage.

Integration Runtime Used: Self-hosted Integration Runtime

Activities:

- Connect to MySQL using SHIR.
- Copy multiple source tables.
- Store each table as CSV or Parquet files in the Raw zone of ADLS.
- No transformations are performed in this pipeline.

*******NOTE:**

Self-hosted Integration Runtime itself "does not allow transformations." Copy Activity can do only limited transformations (such as column mapping or type conversion). **ADF Mapping Data Flows are designed to execute on Azure Integration Runtime (AutoResolve IR or another Azure IR), not on the Self-hosted Integration Runtime.** That's why staging the data in ADLS before running a Mapping Data Flow is the recommended architecture.

Pipeline 2 – Data Transformation (ADLS → Azure SQL Database)

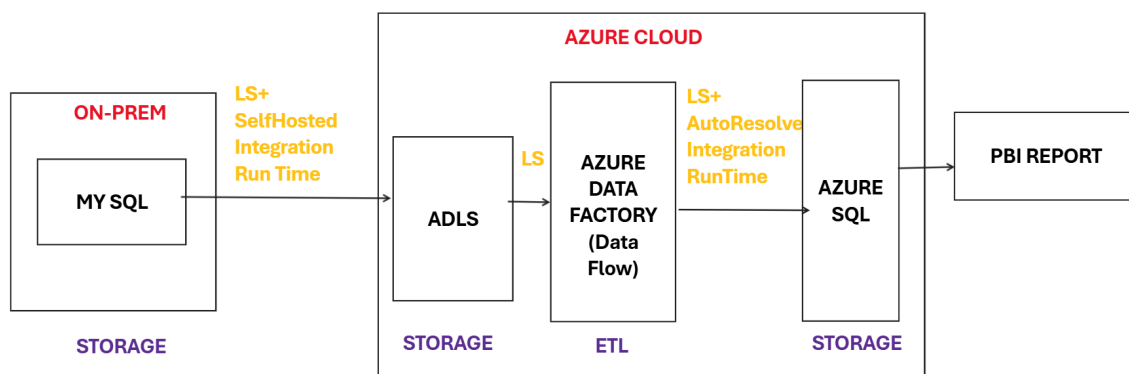
Purpose: Transform the staged data and load it into Azure SQL Database.

Integration Runtime Used: AutoResolve Integration Runtime

Activities:

- Read data from ADLS.
- Perform all required business transformations using Mapping Data Flow.
- Load the transformed data into Azure SQL Database.

ARCHITECTURE:



1)CREATE ADLS:

Home

pipeline6sa_1782453694533 | Overview

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

✓ Your deployment is complete

Deployment name: pipeline6sa_1782453694533

Subscription: NK

Resource group: pipeline6rg

Start time: 26/6/2026, 11:31:52 am

Correlation ID: e4000e62-f6a8-4f0a-9eb6-d2d86de81cbb

Deployment details

Next steps

Go to resource

Give feedback

Tell us about your experience with deployment

Successfully created adls

Home

pipeline6sa_1782453694533 | Overview

pipeline6sa

pipeline6sa | Storage browser

Storage account

Search

Overview

Activity log

Tags

Diagnose and solve problems

Access Control (IAM)

Data migration

Events

Storage browser

Partner solutions

Resource visualizer

Data storage

Security + networking

Data management

Settings

Monitoring

Automation

pipeline6sa

Favorites

Recently viewed

Blob containers

Slogs

data

View all

Classic file shares

Queues

Tables

Add Directory Upload Refresh Delete Copy Paste Rename Acquire lease Break lease Edit columns

Blob containers > data

Authentication method: Access key (Switch to Microsoft Entra user account)

Search blobs by prefix (case-sensitive)

Showing all 0 items

Name	Last modified	Access tier	Blob type	Size
No items found				

2)CREATE AZURE SQL DB:

Home

Microsoft.SQLDatabase.newDatabaseNewServer_0d6da92a640c425eb30a5 | Overview

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

✓ Your deployment is complete

Deployment name : Microsoft.SQLDatabase.newDatabaseNewServer_0d6da92a640c425eb30...

Subscription : NK

Resource group : pipeline6rg

Start time : 26/6/2026, 11:31:52 am

Correlation ID : 19d1e4000e62-f6a8-4f0a-9eb6-d2d86de81cbb

Deployment details

Next steps

Go to resource

Successfully created azure sql db

Home > Microsoft.SqlDatabase.newDatabaseNewServer_0d6da92a640c425eb30a5 | Overview > pipeline6db (pipeline6server/pipeline6db)

pipeline6db (pipeline6server/pipeline6db) | Query editor (preview) ☆ ...

SQL database

Search

New query Templates Open in Feedback Sign out Classic experience

Overview
Activity log
Tags
Diagnose and solve problems
Query editor (preview)
Mirror database in Fabric (preview)
Resource visualizer
Settings
Data management
Integrations
Power Platform
Security
Intelligent performance
Monitoring
Automation
Help

Explorer

Search

pipeline6db
dbo
Queries

Start exploring your data

Write and run SQL queries, create database objects with a preview your data.

New query

No tables were in target

3)CREATE ADF:

Home

Microsoft.DataFactory-20260626113652 | Overview ...

Deployment

Search Delete Cancel Redeploy Download Refresh

Overview
Inputs
Outputs
Template

✓ Your deployment is complete

Deployment name : Microsoft.DataFactory-20260626113652
Subscription : NK
Resource group : pipeline6rg

Start time : 26/6/2026, 11:38:02 am
Correlation ID : 2b2c309e-5ea6-430b-9e2d-86e382218d90

> Deployment details
v Next steps

Go to resource

Give feedback
Tell us about your experience with deployment

Sucessfully created ADF

Data Factory Validate all Publish all

Factory Resources

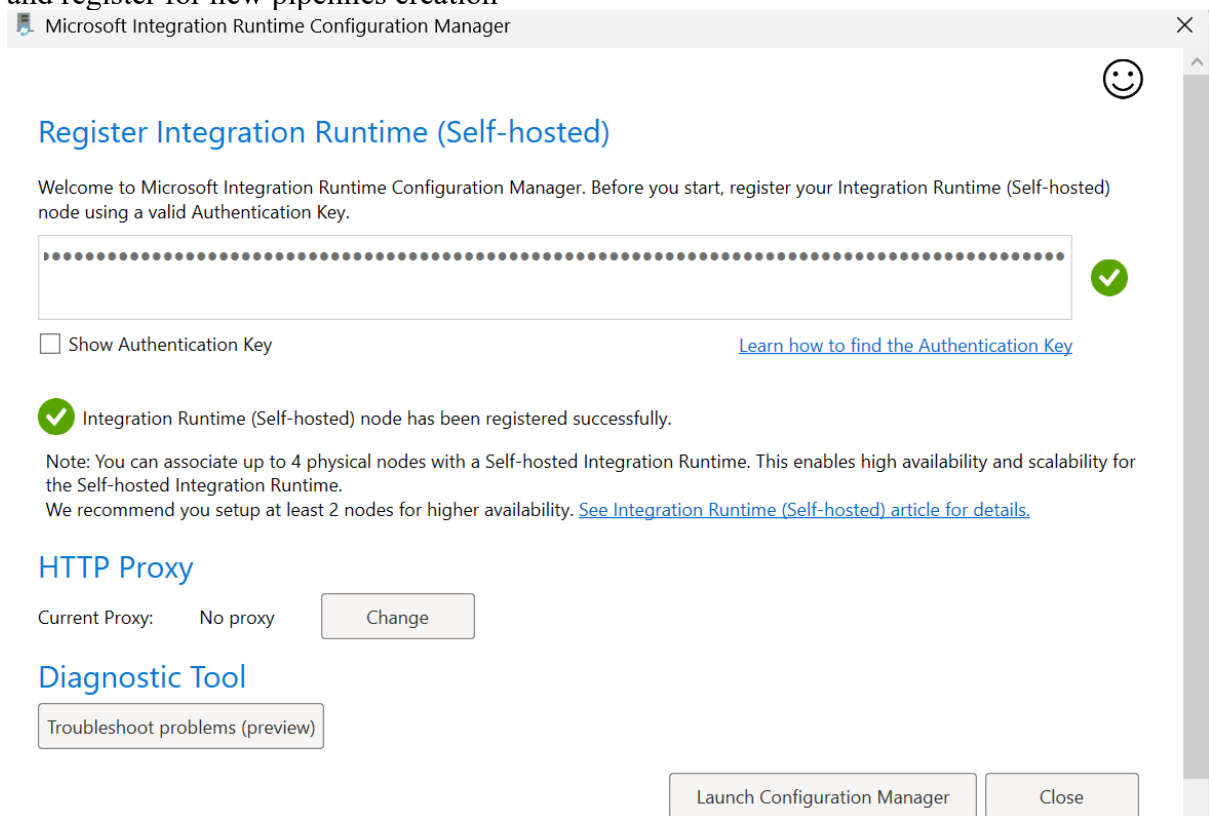
Filter resources by name

Pipelines 0
Change Data Capture (preview) 0
Datasets 0
Data flows 0
Power Query 0

Let's create selfhostedintegrationruntime:

Note:

If selfhostedintegrationruntime---when resource group is deleted---just uninstall and install and register for new pipelines creation



Microsoft Integration Runtime Configuration Manager

Register Integration Runtime (Self-hosted)

Welcome to Microsoft Integration Runtime Configuration Manager. Before you start, register your Integration Runtime (Self-hosted) node using a valid Authentication Key.

.....

☐ Show Authentication Key [Learn how to find the Authentication Key](#)

✓ Integration Runtime (Self-hosted) node has been registered successfully.

Note: You can associate up to 4 physical nodes with a Self-hosted Integration Runtime. This enables high availability and scalability for the Self-hosted Integration Runtime. We recommend you setup at least 2 nodes for higher availability. [See Integration Runtime \(Self-hosted\) article for details.](#)

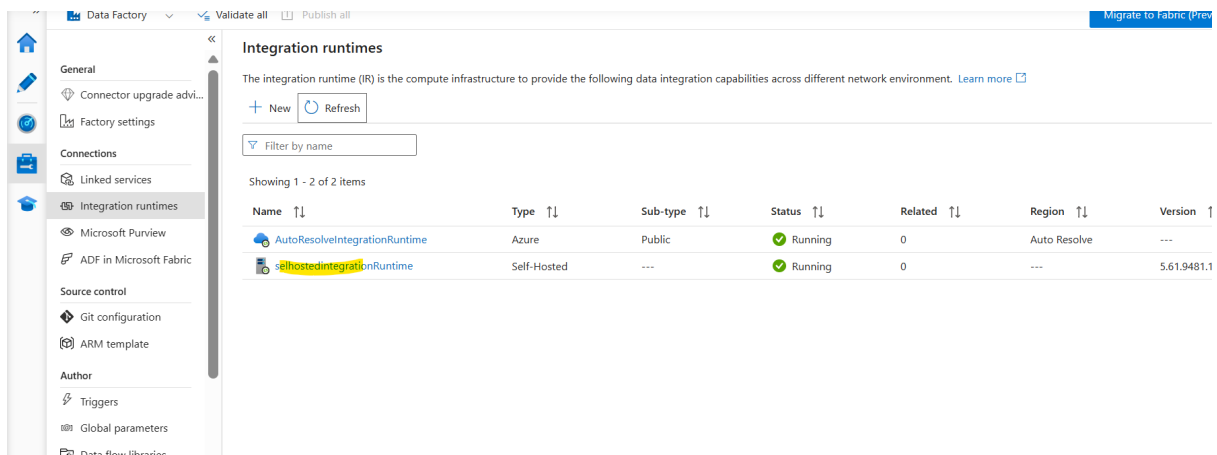
HTTP Proxy

Current Proxy: No proxy Change

Diagnostic Tool

Troubleshoot problems (preview)

Launch Configuration Manager Close



Name	Type	Sub-type	Status	Related	Region	Version
AutoResolveIntegrationRuntime	Azure	Public	Running	0	Auto Resolve	---
selfhostedintegrationruntime	Self-Hosted	---	Running	0	---	5.61.9481.1

Successfully installed and setup the selfhosted integrationruntime

Now create linked service from mysql-adf(create datasets) and then using copy activity – copy the raw data to adls

New dataset

In pipeline activities and data flows, reference a dataset to specify the location and structure of your data within a data store. [Learn more](#)

Select a data store

my

- <AllAzureDatabaseFileGeneric protocol>



Azure Database for MySQL



MySQL

Continue

Cancel

Set properties

Name

Coursedata

Linked service *

MySQL2adf_LS

Connect via integration runtime * ⓘ

✓ selhostedintegrationRuntime

Table name

ksr_courses ('ksr_courses')

☐ Enter manually

OK Back Cancel

>> Data Factory Validate all Publish

Factory Resources

Filter resources by name +

- Pipelines 0
- Change Data Capture (preview) 0
- Datasets 4
 - Batchdata
 - Coursedata
 - Enrollmentdata
 - studentdata
- Data flows 0
- Power Query 0

Co
Lin
Int
Tal

- All the datasets are linked from onprem-adf

- Now we need to copy data to adls

Create a pipeline---copy data---select source(studentdata for example)---sink---+---new---create a linked service from adls2adf---give the path

General
Source
Sink¹
Mapping
Settings
User properties

Source dataset *

Enrollmentdata

Open
+ New
Preview data
Learn more

Use query
☒ Table
☐ Query

Data Factory
Validate all
Publish all

Factory Resources
Filter resources by name
+

Pipelines
1

pipeline1

Dataflows
0

Change Data Capture (preview)

Datasets
5

Batchdata
Coursedata
Enrollmentdata
studentdata
studentdataoutput

Activities
Search activities

Copy data

Data flow

Synapse
Azure Data Explorer
Azure Function
Batch Service
Databricks
General
HDInsight
Iteration & conditionals
Machine Learning
Power Query

Enrollmentdata
Batchdata
Coursedata

Copy data
Copy data

General
Source
Sink¹
Mapping
Settings

Sink dataset *

Select...

Select format
Choose the format type of your data

Avro
Binary
DelimitedText
Iceberg
JSON
ORC

Continue
Back

Set properties

Name

enrollmentdataoutput

Linked service *

Adls2adf_Is

File path

data

Directory

File name

First row as header



Import schema

☒ From connection/store ☐ From sample file ☐ None

> Advanced

OK

Back

Cancel

General

Source

Sink

Mapping

Settings

User properties

Sink dataset *

enrollmentdataoutput

 Open  New [Learn more](#) 

Copy behavior ⓘ

Select...

Max concurrent connections ⓘ

Block size (MB) ⓘ

Metadata ⓘ

 New

Quote all text

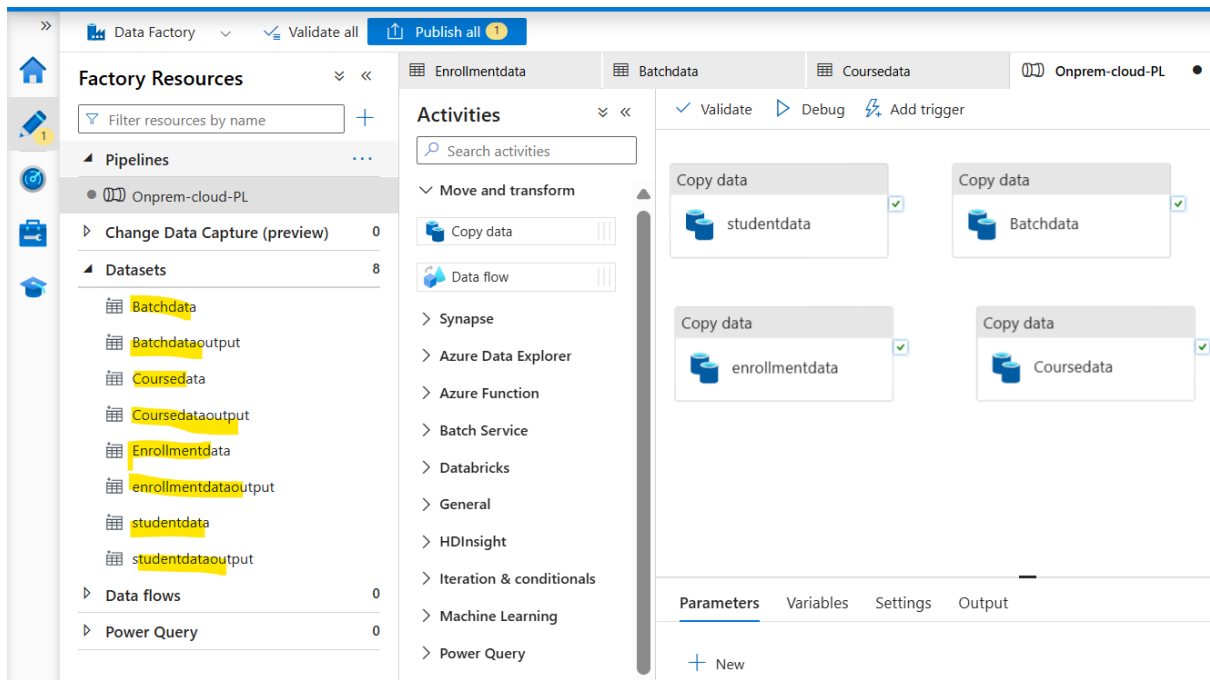


File extension ⓘ

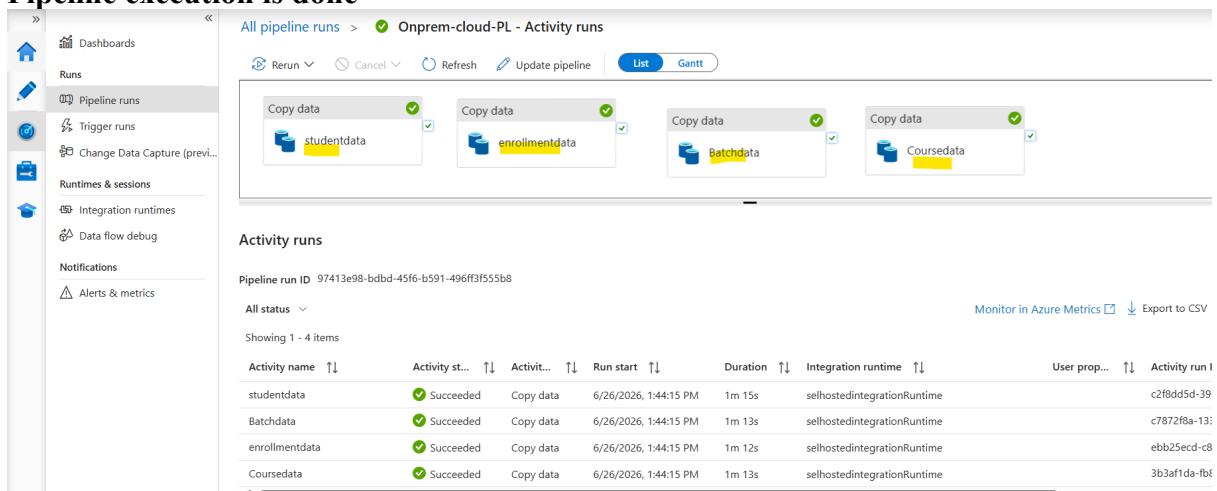
.csv

Max rows per file ⓘ

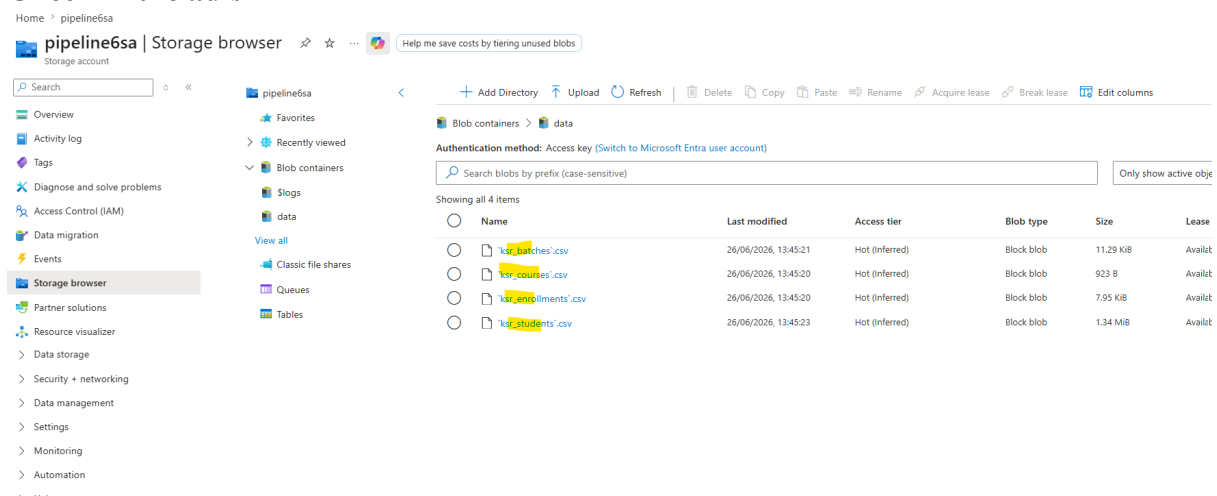
We need to provide the file format here as per requirement



Lets'publish all the changes---add trigger---triggernow---runpipeline---and check the adls whether the data is copied or not
Pipeline execution is done



Check in the adls



Now lets do transformation using adf(dataflow):

We need to bring the data from ads to transform the data

The screenshot displays the Microsoft Fabric Data Factory interface. On the left, the 'Factory Resources' pane shows a tree view with 'Pipelines' (1 item), 'Change Data Capture (preview)' (0 items), 'Datasets' (8 items), 'Data flows' (1 item), and 'Power Query' (0 items). The 'Datasets' list includes 'Batchdata', 'Batchdataoutput', 'Coursedata', 'Coursedataoutput', 'Enrollmentdata', 'enrollmentdataoutput', 'studentdata', and 'studentdataoutput'. The 'Data flows' list includes 'studentenrolldata_DF'. The 'Power Query' list is empty.

The main workspace shows a data flow diagram with a single source named 'source1' with 'Columns: 0 total'. Below the diagram, the 'Source settings' tab is active, showing the following configuration:

- Description:** Add source dataset
- Source type:** Dataset
- Dataset:** Select...
- Options:**
 - ☒ Allow schema drift
 - ☐ Infer drifted column types
 - ☐ Validate schema
- Sampling:** Disable

The 'Set properties' dialog is open, showing the following configuration:

- Name:** studentdatafromads
- Linked service:** Ads2adfs
- File path:** ads / Directory / /var/student1.csv
- First row as header:** ☒
- Import schema:** From connection/store
- Advanced:** (collapsed)

Buttons at the bottom include 'OK', 'Back', and 'Cancel'.

In order to preview the data---make sure the debug mode is on---preview data

Note:

we have 4 tables we need to have 4 sources

Student data is join---enrollmentdata(give join,left table,righttable,common colum)—selectedcolumn---join the final output with---batchdata(give join,left table,righttable,common colum)—selectedcolumn--(give join the final output with---coursedata(give join,left table,right table,common colum)—selectedcolumn---then sink---new-selecttarget(azuresqladb)

Microsoft Azure Data Factory interface showing a pipeline named "studentenrolldata_DF". The pipeline consists of several activities: studentdata, studentenroll..., studentenroll..., studentenrolljoin..., studentenrollbatchs..., studentenrollbatchjo..., and studentenrollbatchcour... The pipeline is currently in the "Validate" state. The left sidebar shows the "Factory Resources" pane with a list of datasets including Batchdata, Enrollmentdata, and Coursedata. The top bar includes "Data Factory", "Validate all", "Publish all", and "Migrate to".

Microsoft Azure Data Factory interface showing the "New dataset" dialog. The dialog is titled "New dataset" and includes a search bar and a list of data stores. The "All" tab is selected, showing a list of data stores including Azure Data Explorer (Kusto), Azure Data Lake Storage Gen2, Azure Database for PostgreSQL, and Azure SQL Database. The "Azure SQL Database" option is highlighted. The dialog also includes a "Continue" button.

New linked service

 Azure SQL Database [Learn more](#) 

 AutoResolveIntegrationKuntime 

Version

☒ 2.0 (Recommended) ☐ 1.0

 Import from connection string

Account selection method

☒ From Azure subscription ☐ Enter manually

Azure subscription

NK (5bd8f1cd-03ad-48fa-8410-6e58274fc11f) 

Server name *

pipeline6server  

Database name *

pipeline6db  

Authentication type *

SQL authentication 

User name *

adminnk

Password

[Azure Key Vault](#)

Password *

Create

Cancel

 Connection successful

 Test connection

Set properties

Name

studentcoursedata

Linked service *

adf2azsqldb

☐ Select from existing table ☒ New table

Schema and table name

Schema name

studentcourse_TB

> Advanced

OK

Back

Cancel

Publish all the changes and add the dataflow in the pipeline---add trigger---triggernow-run pipeline

The screenshot displays the Azure Data Factory (ADF) console. On the left, the 'Factory Resources' pane shows a list of datasets and data flows. The 'Data flows' section is expanded, showing 'studentenrolldata_DF'. The main workspace shows the 'Copy data' activity configuration for 'studentenrolldata_DF'. The 'Settings' tab is active, showing the 'Data flow' dropdown set to 'studentenrolldata_DF', 'Run on (Azure IR)' set to 'AutoResolveIntegrationRuntime', and 'Compute size' set to 'Small'. The 'Logging level' is set to 'Verbose'. The 'Activities' pane on the left shows the 'Copy data' activity selected. The 'Data flow' configuration is shown in a preview window on the right, displaying a data flow graph with a single activity named 'Data flow1'.

Add trigger-triggernow
Successfully pipeline run

Activity runs

Pipeline run ID: 5251d922-d9af-4c5f-a248-496b7fe0ef94

All status: Monitor in Azure Metrics Export to CSV

Showing 1 - 5 items

Activity name	Activity status	Activity type	Run start	Duration	Integration runtime	User properties	Activity run ID	Location
enrollmentdata	Succeeded	Copy data	6/26/2026, 2:40:37 PM	1m 19s	selfhostedintegrationRuntime		10b1e34a-2229-4967-a19b-be0bd5d09cde	
Data flow1	Succeeded	Data flow	6/26/2026, 2:40:37 PM	5m 6s	AutoResolveIntegrationRuntime (East US)		e9252d68-a37b-4ba2-927e-484b56ce97f6	
studentdata	Succeeded	Copy data	6/26/2026, 2:40:37 PM	1m 10s	selfhostedintegrationRuntime		d74982e2-160e-4b43-9432-18fca220691e	
Batchdata	Succeeded	Copy data	6/26/2026, 2:40:37 PM	1m 9s	selfhostedintegrationRuntime		004a250e-2f42-41b4-a331-d47d03bc2bd5	
Coursedata	Succeeded	Copy data	6/26/2026, 2:40:37 PM	1m 8s	selfhostedintegrationRuntime		2f3fbeb5-ab71-41bf-9048-13e8912c6947	

Check in ads and target

pipeline6db (pipeline6server/pipeline6db) | Query editor (preview)

SQL database

Search

New query Templates Open in Feedback Sign out Classic experience

Overview Activity log Tags Diagnose and solve problems Query editor (preview) Mirror database in Fabric (preview) Resource visualizer Settings Data management Integrations Power Platform Security Intelligent performance Monitoring Automation Help

Explorer

pipeline6db

dbo

Tables

studentcourse_

Columns

Indexes

Views

Stored Procedures

Functions

Queries

SQL query 1

SQL query 1

Run Save as view

1 SELECT TOP (1000) * FROM [dbo].[studentcourse_TB]

Messages Results ()

student_id	first_name	last_name	join_date	batch_id	course_id	course_name
140	Keerthi	Sekhar	2025-04-02	74	10	Cloud Fundamentals (AZ...
171	Rohit	Kumar	2024-05-01	38	8	SQL Zero to Hero
214	Sanjana	Gopal	2024-06-29	28	6	Practical Machine Learn...
222	Meera	Sekhar	2023-11-05	110	7	GenAI for Data Engineers
283	Harini	Roy	2024-06-24	39	10	Cloud Fundamentals (AZ...
295	Harini	Pillai	2024-10-11	103	9	Python for Data Enginee...
342	Rohit	Gowda	2024-11-25	25	6	Practical Machine Learn...
351	Ajgun	Kumar	2023-03-02	26	3	AWS Data Engineering w...
388	Aishwarya	Nambiar	2025-07-08	72	6	Practical Machine Learn...

Succeeded [0 sec 408 ms] Columns: 7 Rows: 200

*****DRAWBACKS:

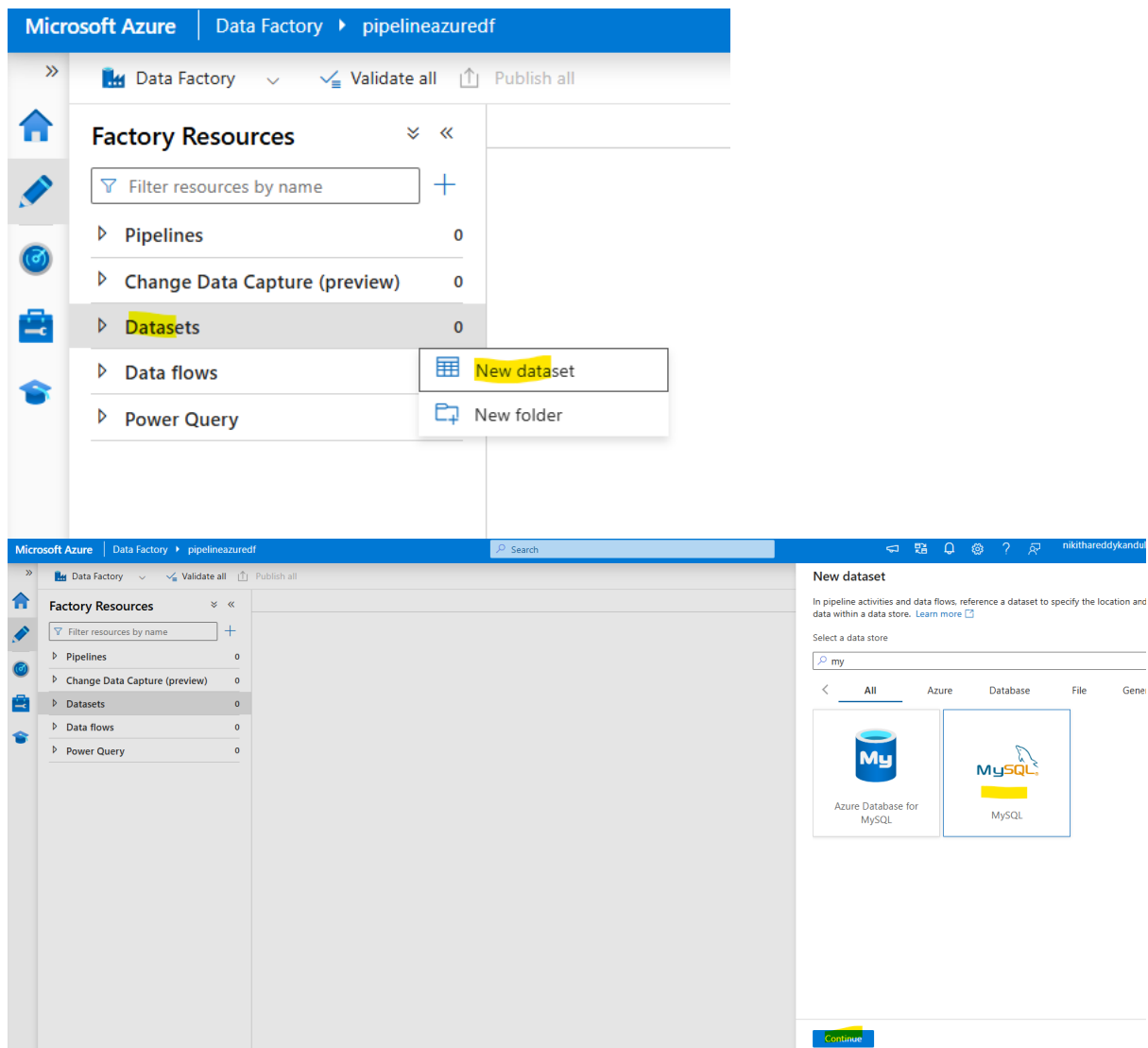
The main drawback of this method is we have to create multiple datasets for example if we have 50 tables need to create 100 tables(50 for inputs and 50 for output)

*****Method-2: Using Parameters in generalized datasets:

Problem with method 1: example if we have 50 tables need to create 100 tables(50 for inputs and 50 for output)

Solution: Creating parameters with generalized tables

Create Generalised Dataset:



Create a linked service:

New linked service

 MySQL [Learn more](#) 

Name *

MySQL2adf

Description

Connect via integration runtime * 

 selfhostedintegrationRuntime1 

 The credentials are stored in the machines of self-hosted integration runtime if you don't choose to store them in Azure Key Vault.

Server name *

localhost

Port

3306

Database name *

ksr_db1

User name *

root

Password

Azure Key Vault

Password *

 Connection successful

 Test connection

Create

Cancel

Hence it is a generalized table no need to select any table

Set properties

Name

InputData

Linked service *

MySQL2adf

Connect via integration runtime * ⓘ

✓ selfhostedintegrationRuntime1

Table name

Loading...

☐ Enter manually

OK

Back

Cancel

Now create input parameters

The screenshot shows the 'Set properties' dialog for a dataset named 'InputData'. The 'Name' is 'InputData', the 'Linked service' is 'MySQL2adf', and the 'Connect via integration runtime' is 'selfhostedintegrationRuntime1'. The 'Table name' is 'Loading...' and the 'Enter manually' checkbox is unchecked. Below the dialog, the 'Parameters' tab of the 'InputData' dataset configuration is shown. The 'Factory Resources' pane on the left lists 'Pipelines' (0), 'Change Data Capture (preview)' (0), 'Datasets' (1), 'Data flows' (0), and 'Power Query' (0). The 'InputData' dataset is selected under 'Datasets'. The 'Parameters' tab shows the 'Linked service' as 'MySQL2adf', the 'Integration runtime' as 'selfhostedintegrationRuntime1', and the 'Table' as 'Loading...'. The 'Enter manually' checkbox is unchecked. The 'Test connection' button is disabled, and the 'Edit' button is active. The 'New' button is also active. The 'Learn more' link is present.

» Data Factory Validate all Publish all 1

Factory Resources

Filter resources by name

- Pipelines 0
- Change Data Capture (preview) 0
- Datasets 1**
 - InputData
- Data flows 0
- Power Query 0

InputData

MySQL InputData

Connection Parameters

+ New Delete

Name	Type	Default value
Inputdatatable	String	Value

Create the generalized output data :

New dataset

In pipeline activities and data flows, reference a dataset to specify the location and structure of your data within a data store. [Learn more](#)

Select a data store

Search

< All Azure Database File Generic protocol >

Azure Cosmos DB for NoSQL	Azure Data Explorer (Kusto)	Azure Data Lake Storage Gen2
		
Azure Database for MySQL	Azure Database for PostgreSQL	Azure Databricks Delta Lake
		
Azure File Storage	Azure SQL Database	Azure SQL Database Managed Instance

Continue Cancel

New linked service

 Azure SQL Database [Learn more](#) 

 AutoResolveIntegrationRuntime 

Version (Recommended)

☒ 2.0 (Recommended) ☐ 1.0

 Import from connection string

Account selection method ?

☒ From Azure subscription ☐ Enter manually

Azure subscription

NK (5bd8f1cd-03ad-48fa-8410-6e58274fc11f) 

Server name *

pipelinesa 



Database name *

pipelinedb 



Authentication type *

SQL authentication 

User name *

adminnk

Password

Azure Key Vault

Password *

Create

Cancel

 Connection successful

 Test connection

Set properties

Name

OutputData

Linked service *

adf2AzureSqlDb_Is

Table name

Select...

☐ Enter manually

Import schema

☐ From connection/store ☒ None

OK

Back

Cancel

Create a pipeline to copy the data

The screenshot displays the Microsoft Azure Data Factory portal interface. The top navigation bar shows 'Microsoft Azure | Data Factory | pipelineazuredf'. The left sidebar contains 'Factory Resources' with a search bar and a list of resources: Pipelines (1), Datasets (2), Data flows (0), and Power Query (0). The 'Pipelines' section is expanded, showing a pipeline named 'Onprem2cloud_PL'. The 'Activities' section is also expanded, showing a list of activities: Move and transform, Synapse, Azure Data Explorer, Azure Function, Batch Service, Databricks, General, HDInsight, Iteration & conditionals, Machine Learning, and Power Query. The 'Copy data' activity is selected, and its configuration is shown in the main pane. The 'Copy data' activity is named 'Copy data1' and is configured with the following properties:

- Name: Copy data1
- Description: (empty)
- Activity state: ☒ Activated ☐ Deactivated
- Timeout: 0.12:00:00
- Retry: 0

The 'Source' tab is selected, showing the 'Copy data' activity. The 'Sink' tab is also visible, showing the 'Copy data' activity. The 'Mapping' tab is also visible, showing the 'Copy data' activity. The 'Settings' tab is also visible, showing the 'Copy data' activity. The 'User properties' tab is also visible, showing the 'Copy data' activity.

General **Source** Sink¹ Mapping Settings User properties

Source dataset * InputData

[Open](#) [+ New](#) [Preview data](#) [Learn more](#)

Dataset properties ⓘ

Name	Value	Type
Inputdatatable	Value Add dynamic content [Alt+Shift+D]	sti

Use query ☒ Table ☐ Query

Additional columns ⓘ [+ New](#)

In the place of value we can give the table name

General **Source** Sink¹ Mapping Settings User properties

Source dataset * InputData

[Open](#) [+ New](#) [Preview data](#) [Learn more](#)

Dataset properties ⓘ

Name	Value
Inputdatatable	ksr_attendance

Use query ☒ Table ☐ Query

Additional columns ⓘ [+ New](#)

Note:

Whatever the value I gave in the source of input table---this is getting from mysql(onprem)---table

▲ Datasets 2

- InputData
- OutputData

▶ Data flows 0

▶ Power Query 0

Connection Parameters

Linked service * MySQL2adf [Test connection](#)

Integration runtime * selfhostedintegrationRuntime1 [Edit](#)

Table @dataset().Inputdatatable [Preview data](#)

Activities

Search activities

Move and transform

Copy data

Data flow

Synapse

Azure Data Explorer

Azure Function

Batch Service

Databricks

General

HDInsight

Iteration & conditionals

Machine Learning

Power Query

Validate Validate copy runtime Debug Add trigger

Copy data

Copy data1

General Source **Sink** Mapping Settings User properties

Sink dataset * OutputData Open +

Write behavior ☒ Insert ☐ Upsert ☐ Stored procedure

Bulk insert table lock ① ☐ Yes ☒ No

Table option ☐ Use existing ☒ Auto create table ①

Pre-copy script ①

Write batch timeout ① e.g. 00:30:00

Write batch size ①

Max concurrent connections ①

Now for output table we need to give the table name

Factory Resources

Filter resources by name +

Pipelines 1

Onprem2cloud_PL

Change Data Capture (preview) 0

Datasets 2

InputData

OutputData

Data flows 0

Power Query 0

Activities

Search activities

Move and transform

Copy data

Data flow

Synapse

Azure Data Explorer

Azure Function

Batch Service

Databricks

General

HDInsight

Iteration & conditionals

Machine Learning

Power Query

Validate Validate copy runtime Debug Add trigger

Copy data

Copy data1

General Source **Sink** Mapping Settings User properties

Sink dataset * OutputData Open + New Learn more

Dataset properties ①

Name	Value
OutputdataTable	Attendance_TB

Write behavior ☒ Insert ☐ Upsert ☐ Stored procedure

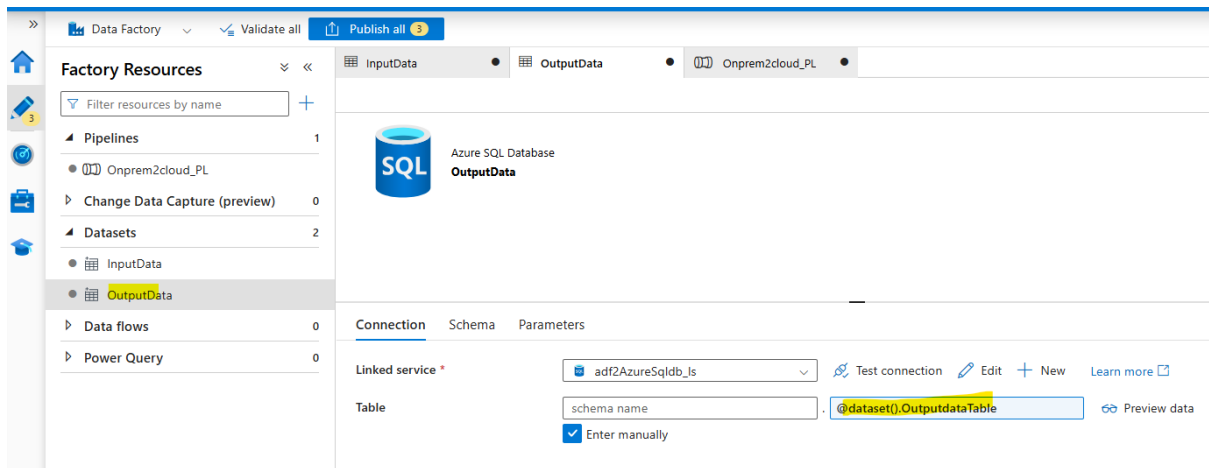
Bulk insert table lock ① ☐ Yes ☒ No

Table option ☐ Use existing ☒ Auto create table ①

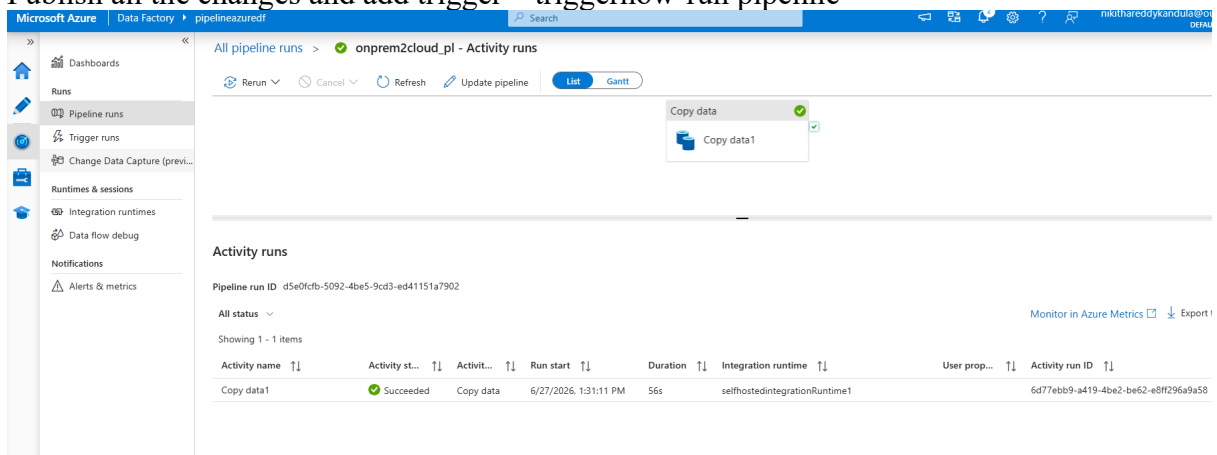
Pre-copy script ①

NOTE:

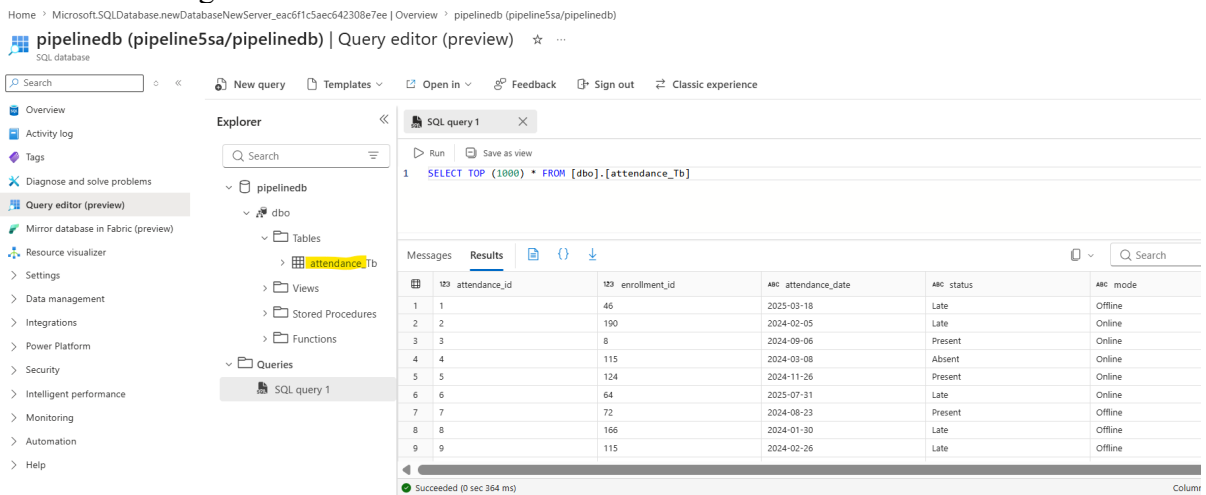
Whatever the table name I gave in the output table will be written in target (Azure sql db)



Publish all the changes and add trigger—triggernow-run pipeline



Check in the target



For example if I need exams table

Go to update pipeline---give the table name in source(mysql table) ---give the table name to write in the output data(value)

Microsoft Azure | Data Factory > pipelineazuredf

Search

Dashboards

Runs

Pipeline runs

Trigger runs

Change Data Capture (previ...

Runtimes & sessions

Integration runtimes

Data flow debug

Notifications

Alerts & metrics

All pipeline runs > onprem2cloud_pl - Activity runs

Rerun

Cancel

Refresh

Update pipeline

List

Gantt

Copy data

Copy data1

Activity runs

Pipeline run ID d5e0fcfb-5092-4be5-9cd3-ed41151a7902

All status

Showing 1 - 1 items

Activity name	Activity st...	Activit...	Run start	Duration	Integration runtime
Copy data1	Succeeded	Copy data	6/27/2026, 1:31:11 PM	56s	selfhostedintegration

Data Factory | Validate all | Publish all

Factory Resources

Filter resources by name

Pipelines

onprem2cloud_pl

Change Data Capture (preview)

Datasets

InputData

OutputData

Data flows

Power Query

Activities

Search activities

Move and transform

Synapse

Azure Data Explorer

Azure Function

Batch Service

Databricks

General

HDInsight

Iteration & conditionals

Machine Learning

Power Query

InputData

OutputData

onprem2cloud_pl

Validate

Validate copy runtime

Debug

Add trigger

Copy data

Copy data1

General

Source

Sink

Mapping

Settings

User properties

Source dataset * InputData

Dataset properties

Name

Value

Inputdatatable

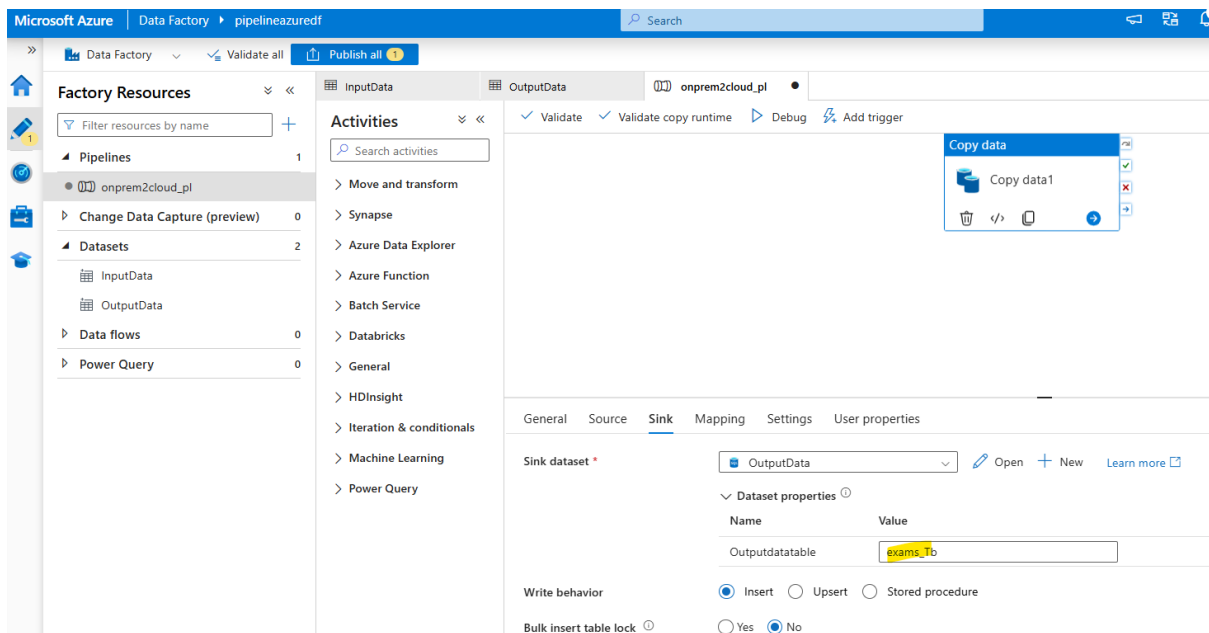
ksr_exam

Add dynamic content [Alt+Shift+D]

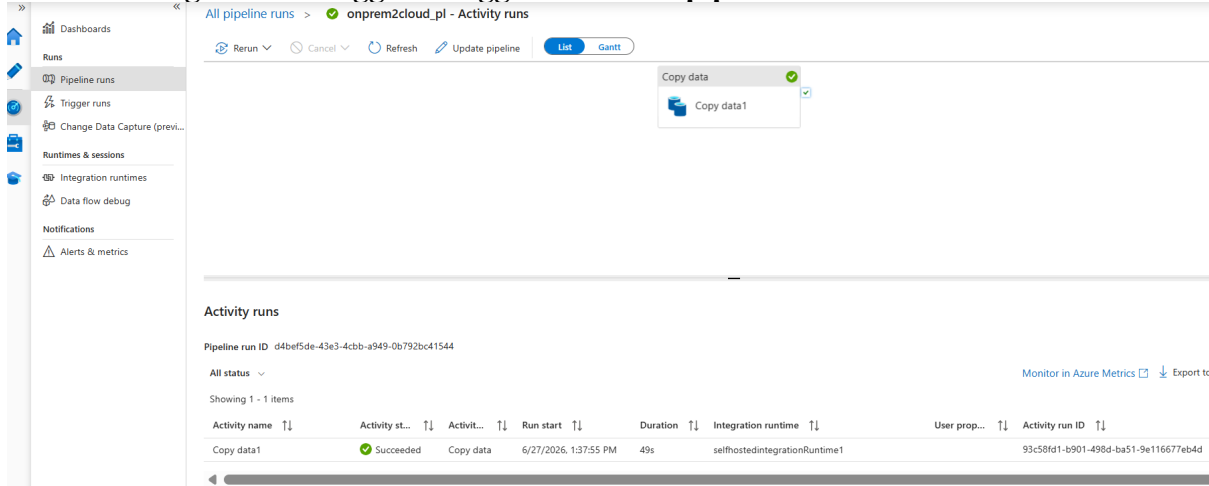
Use query

Table

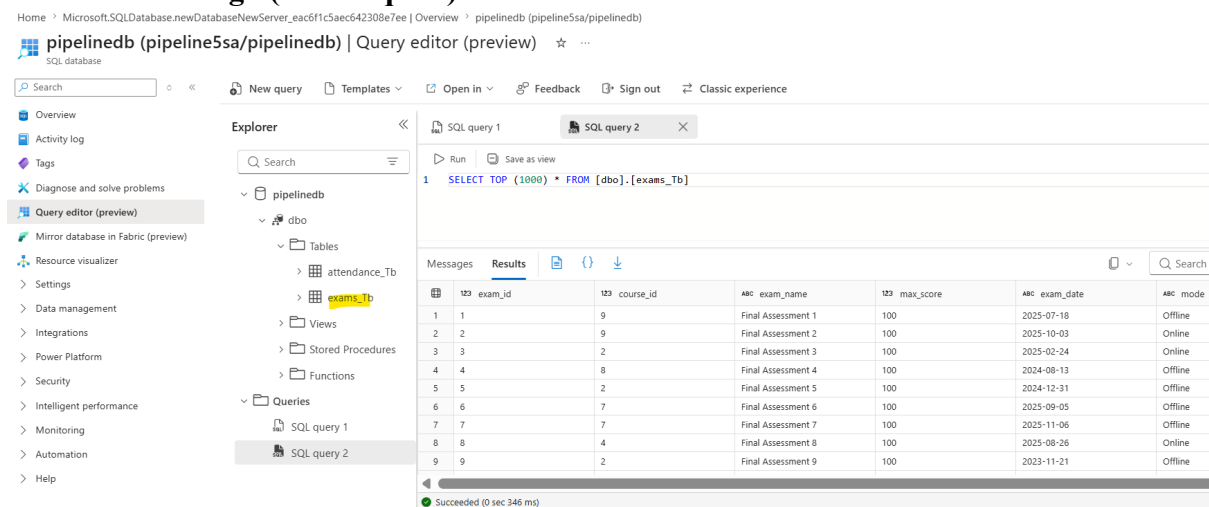
Query



Publish changes---add trigger---triggernow—run pipeline



Check in the target(azure sql db)



***Drawback:

- Eventhough the multiple datasets creation problem is resolved with the help of paarmeters but still we have publishing issues (if 50 tables need to copy we need to publish it 50 time)

- even created the parameters in datasets hardcoded within the pipeline to give the source and target tables

*****METHOD-3:Pipeline level parameters

Problem:

- Hardcoded with the table names in source and target with in the pipeline
- Publishing issue issues (if 50 tables need to copy we need to publish it 50 time)

Solution:

- Creating the pipeline level parameters

Create input and output generalized datasets: donot give the table names while creating a linked service

The image shows two screenshots of the Azure Data Factory (ADF) interface, illustrating the configuration of a pipeline-level parameter for a dataset.

Top Screenshot: The 'Factory Resources' pane on the left shows the 'inputdata' dataset selected. The main pane displays the 'Parameters' tab for the 'inputdata' dataset. A table is shown with the following structure:

Name	Type	Default value
inputsourceTable	String	Value

Bottom Screenshot: The 'Factory Resources' pane on the left shows the 'inputdata' dataset selected. The main pane displays the 'Connection' tab for the 'inputdata' dataset. The configuration is as follows:

Property	Value	Actions
Linked service *	MySQL1_ls	Test connection, Edit, + New, Learn more
Integration runtime *	selfhostedintegrationRuntime1	Edit
Table	@dataset().inputsourceTable	Preview data

Factory Resources

- Pipelines: 0
- Change Data Capture (preview): 0
- Datasets: 2
 - inputdata
 - ouputdata
- Data flows: 0
- Power Query: 0

Parameters

Name	Type	Default value
outputtarget	String	Value

Connection

Linked service: **adf2AzureSqlDatabase1_ls** (Test connection, Edit, New, Learn more)

Table: **schema name** (Enter manually) | **@dataset().outputtarget** (Preview data)

Create a pipeline level parameters

Factory Resources

- Pipelines: 1
 - pipeline1
- Change Data Capture (preview): 0
- Datasets: 2
 - inputdata
 - ouputdata
- Data flows: 0
- Power Query: 0

Activities

- Move and transform
 - Copy data
 - Data flow
- Synapse
 - Azure Data Explorer
 - Azure Function
 - Batch Service
 - Databricks
 - General
 - HDInsight
 - Iteration & conditionals
 - Machine Learning
 - Power Query

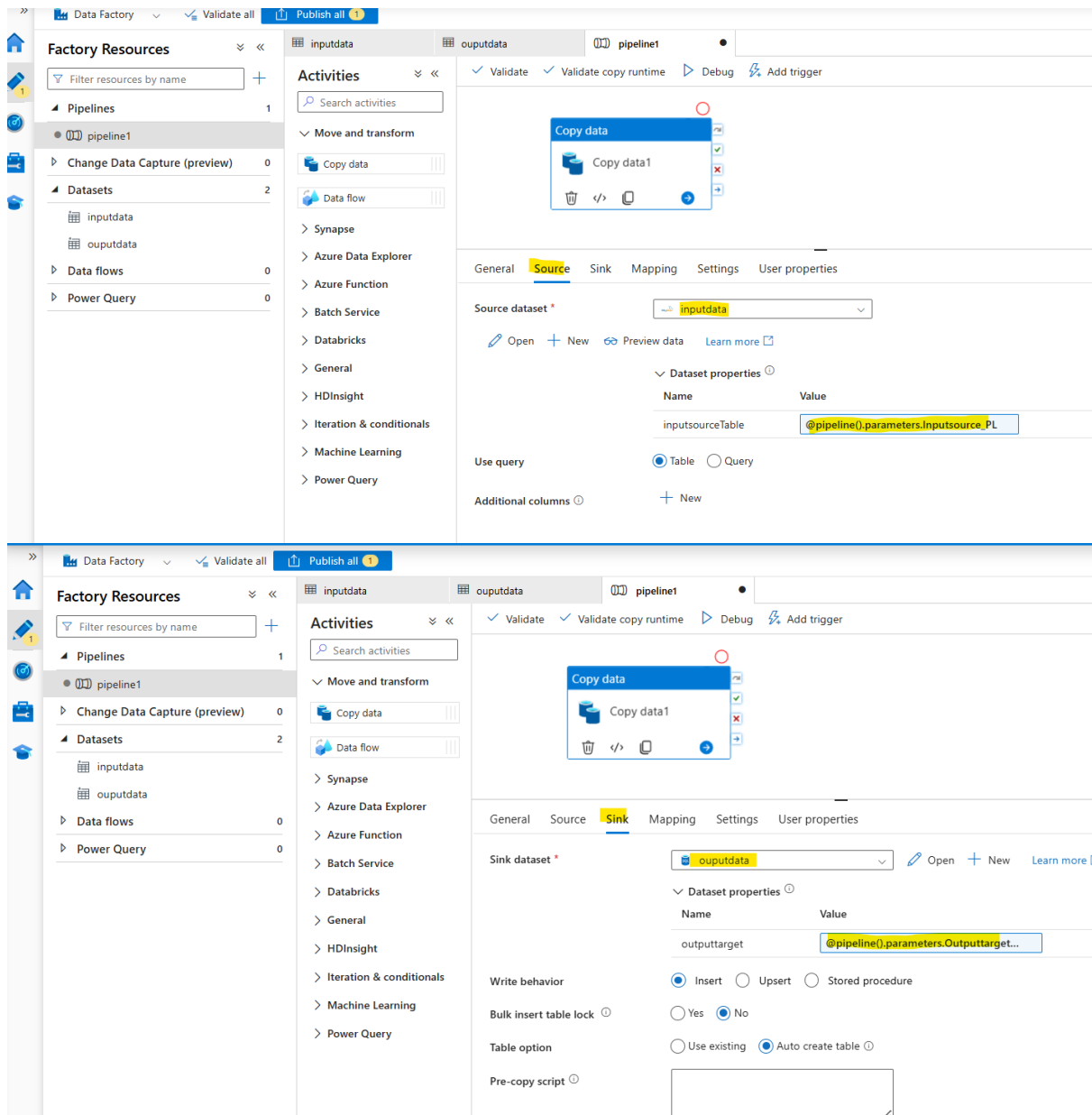
Copy data

Copy data1

Parameters

Name	Type	Default value
Inputsource_PL	String	Value
Outputtarget_PL	String	Value

Now in the pipeline we are not giving the source and target table we are giving the parameters created for input and output



Publish---addtrigger---triggernow---now we need to give the source and target tables---ok---to run pipeline

Pipeline run

⚠ Trigger pipeline now using last published configuration.

Parameters
of the last

Name	Type	Value
Inputsource_PL	string	ksr_batches
Outputtarget_PL	string	batches_Tb

OK Cancel

Microsoft Azure

Data Factory

pipelineazuredf

Search

»

«

Home

Dashboard

Runs

Pipeline runs

Trigger runs

Change Data Capture (previ...

Runtimes & sessions

Integration runtimes

Data flow debug

Notifications

Alerts & metrics

All pipeline runs > ✓ pipeline1 - Activity runs

Rerun

Cancel

Refresh

Update pipeline

List

Gantt

Copy data

✓

Copy data1

✓

Activity runs

Pipeline run ID 54b5f252-90e3-4379-84e6-5937cdea4b75

All status

Showing 1 - 1 items

Activity name	Activity st...	Activity type	Run start	Duration	Integration runtime
Copy data1	✓ Succeeded	Copy data	6/27/2026, 2:04:38 PM	40s	selfhostedintegrationRuntime1

Check in target



Search

Overview

Activity log

Tags

Diagnose and solve problems

Query editor (preview)

Mirror database in Fabric (preview)

Resource visualizer

Settings

Data management

Integrations

Power Platform

Security

Intelligent performance

Monitoring

Automation

Help

Explorer

Search

pipelinedb

dbo

Tables

batches_Tb

Views

Stored Proced...

Functions

Queries

SQL query 1

SQL query 2

SQL query 3

SQL query 4

SQL query 1SQL query 2SQL query 3SQL query 4SQL query 5

RunSave as view

1SELECT TOP (1000) * FROM [dbo].[batches_Tb]

MessagesResults

	batch_id	course_id	trainer_id	ABC_batch_code	ABC_mode	ABC_slot	ABC_start_date	ABC_en
1	1	3	4	B2025001	Offline	Weekday Morning	2025-01-05	2025-0
2	2	5	5	B2024002	Hybrid	Weekday Evening	2024-04-19	2024-0
3	3	4	39	B2024003	Offline	Weekday Morning	2024-09-05	2024-1
4	4	4	50	B2024004	Hybrid	Weekend Morning	2024-08-09	2024-1
5	5	9	29	B2024005	Online	Weekday Morning	2024-04-17	2024-0
6	6	1	22	B2024006	Hybrid	Weekend Morning	2024-03-22	2024-0
7	7	8	44	B2025007	Offline	Weekend Evening	2025-01-02	2025-0
8	8	3	35	B2023008	Hybrid	Weekday Morning	2023-12-25	2024-0
9	9	7	17	B2024009	Online	Weekend Morning	2024-12-30	2025-0

Succeeded (0 sec 474 ms)

For example I need feedback table
Go to updatepipeline---addtrigger---triggernow---give source and target table---ok-run pipeline

Pipeline run

Trigger pipeline now using last published configuration.

Parameters

Name	Type	Value
Inputsource_PL	string	ksr_feedback
Outputtarget_PI	string	feedback_tb

OK Cancel

Activity runs

Pipeline run ID f5cde148-a1fb-4b62-80fb-d4d7099f7eb1

All status ▾

Showing 1 - 1 items

Activity name	Activity st...	Activit...	Run start	Duration	Integration runtime	User prop...	Activity run ID
Copy data1	✓ Succeeded	Copy data	6/27/2026, 2:07:59 PM	41s	selfhostedIntegrationRuntime1		bad9afc0-0d73-4608-91db-943ce67

Home > Microsoft.SqlDatabase.newDatabaseNewServer_eac6f1c5aec642308e7ee | Overview > pipelinedb (pipeline5sa/pipelinedb)

pipelinedb (pipeline5sa/pipelinedb) | Query editor (preview) ☆ ...

SQL database

Search

New query Templates Open in Feedback Sign out Classic experience

Explorer

Search

pipelinedb

dbo

Tables

feedback_tb

Views

Stored Proced...

Functions

Queries

SQL query 1

SQL query 2

SQL query 3

SQL query 1

SQL query 2

SQL query 3

SQL query 4

SQL query 5

Run Save as view

1 SELECT TOP (1000) * FROM [dbo].[feedback_tb]

Messages Results {}

	123 feedback_id	123 student_id	123 course_id	123 rating	ABC comments
1	1	5831	5	2	Pace was slightly fast but cont...
2	2	4573	2	3	Course material could be mor...
3	3	6379	6	1	More interview preparation w...
4	4	5014	10	4	Need more focus on fundame...
5	5	1914	8	4	More interview preparation w...

✓ Succeeded (0 sec 980 ms)

****Drawback:

- Pipeline level we are hard coding the table names
- If we want to copy 100 tables we need to hardcode 100 times

*****Method-4: BEST METHOD: (Real time projects):

Main Objective:

I want to copy all the tables for the ksr_db1 database where I want to use dynamic way ---in future if a new table comes in I need to fetch it

Create a pipeline:

***** select * from INFORMATION_SCHEMA.TABLES -----this query gives all the system tables in mysql

From this I want to read the tables from ksr_db1

select * from INFORMATION_SCHEMA.TABLES WHERE TABLE_SCHEMA='KSR_db1';

From this I want to read the only all tables names from ksr_db1

select Table_name from INFORMATION_SCHEMA.TABLES WHERE TABLE_SCHEMA='KSR_db1';

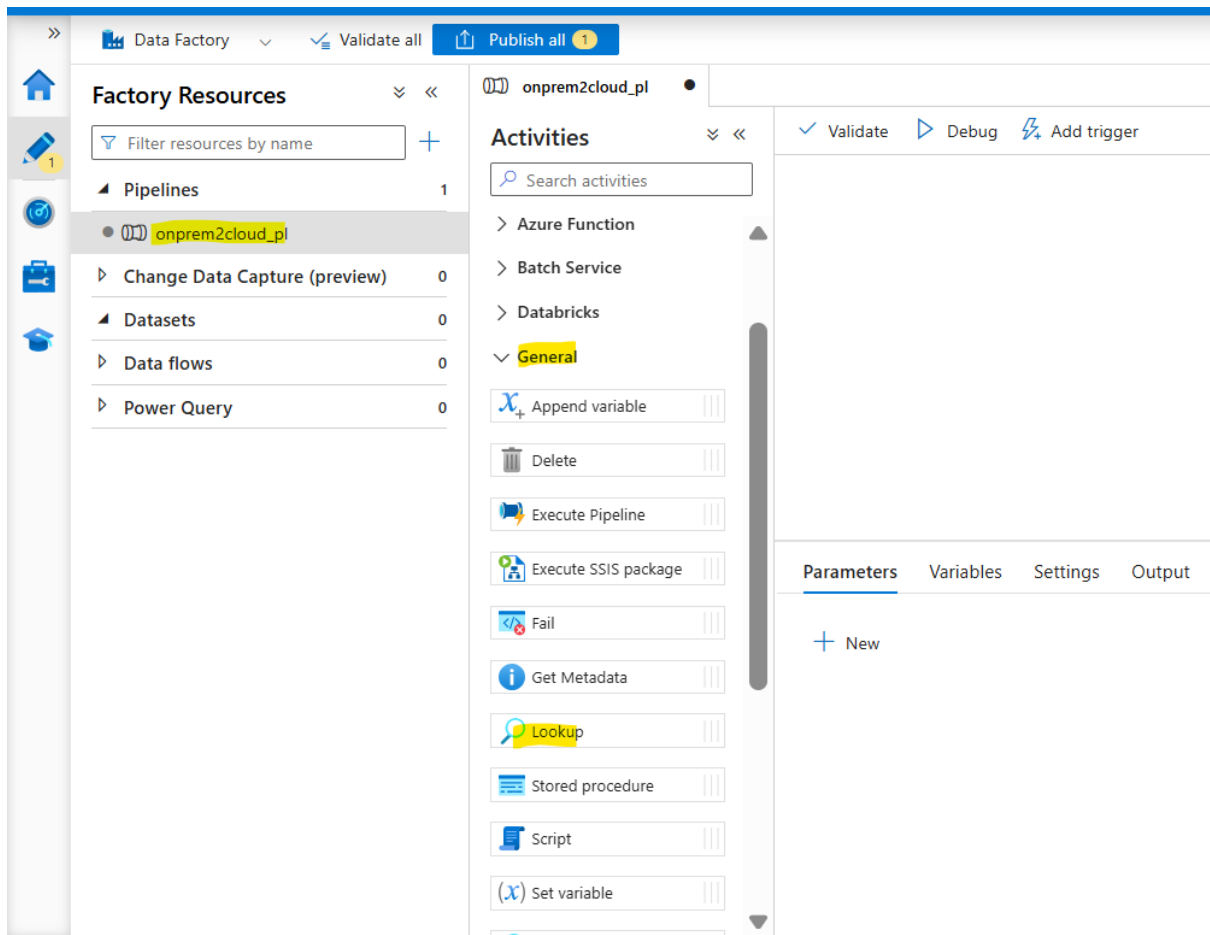

```
48 • select Table_name from INFORMATION_SCHEMA.TABLES WHERE TABLE_SCHEMA='KSR_db1';
49
50
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
TABLE_NAME				
ksr_attendance				
ksr_batches				
ksr_courses				
ksr_enrollments				
ksr_exam_results				
ksr_exams				
ksr_feedback				
ksr_payments				
ksr_students				
ksr_trainers				

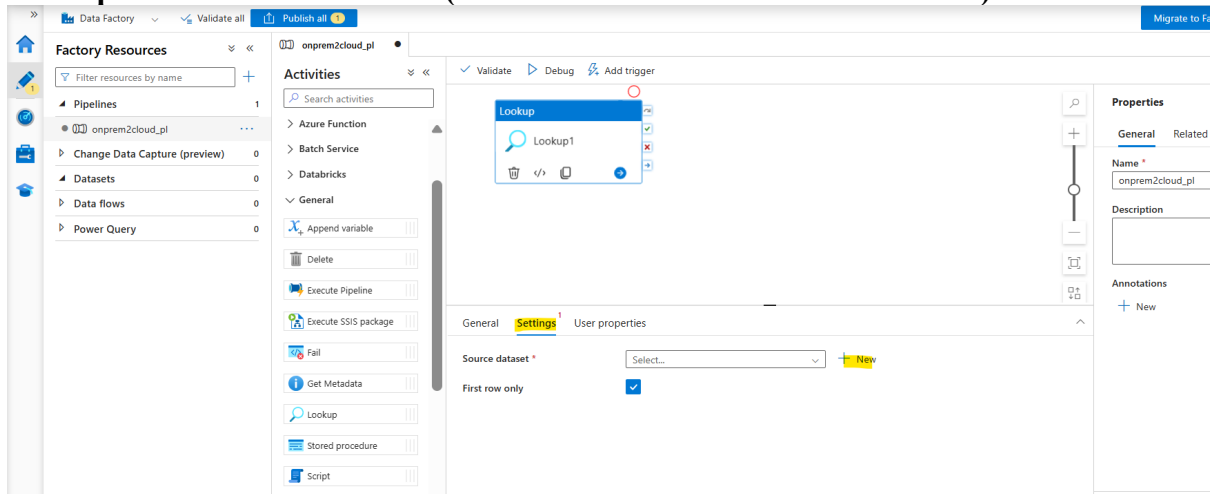
Now I want to read all the tables from the above

```
48 • select Table_name from INFORMATION_SCHEMA.TABLES WHERE TABLE_SCHEMA='KSR_db1';
49
50
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
TABLE_NAME				
ksr_attendance				
ksr_batches				
ksr_courses				
ksr_enrollments				
ksr_exam_results				
ksr_exams				
ksr_feedback				
ksr_payments				
ksr_students				
ksr_trainers				



Lookup is used to read the table(lets create new table to refer the tables)



Factory Resources

- Pipelines: 1
 - onprem2cloud_pl
- Change Data Capture (preview): 0
- Datasets: 0
- Data flows: 0
- Power Query: 0

Activities

- Azure Function
- Batch Service
- Databricks
- General
 - Append variable
 - Delete
 - Execute Pipeline
 - Execute SSIS package
 - Fail
 - Get Metadata
 - Lookup
 - Stored procedure
 - Script
 - Set variable

Lookup1

Source dataset: Select...

First row only: ☒

New dataset

In pipeline activities and data flows, reference a dataset to specify the location and structure of your data within a data store. [Learn more](#)

Select a data store

my

All Azure Database for MySQL MySQL

Set properties

Name: ReferenceTables

Linked service: MySQLIs

Connect via integration runtime: ☒ Azure Integration runtime1

Table name: Select...

☐ Enter manually

[Advanced](#)

OK Back

Enable the query

Factory Resources

- Pipelines: 1
 - onprem2cloud_pl
- Change Data Capture (preview): 0
- Datasets: 1
 - ReferenceTables
- Data flows: 0
- Power Query: 0

Activities

- Azure Function
- Batch Service
- Databricks
- General
 - Append variable
 - Delete
 - Execute Pipeline
 - Execute SSIS package
 - Fail
 - Get Metadata
 - Lookup
 - Stored procedure
 - Script
 - Set variable

Lookup1

Source dataset: ReferenceTables

First row only: ☒

Use query: ☒ Query

Query:

Edit

Write the query to read the tables from the db

Factory Resources

onprem2cloud_pl

Activities

Lookup

Settings

Source dataset * ReferenceTables

First row only ☐

Use query ☐ Table ☒ Query

Query * `select Table_name from INFORMATION_SCHEMA.TABLES WHERE TABLE_SCHEMA='KSR_dB1';`

Now publish all changes---add trigger---trigger now---run pipeline
Successfully ran the pipeline

onprem2cloud_pl - Activity runs

Run Succeeded

Successfully ran onprem2cloud_pl (Pipeline).

Activity runs

Pipeline run ID: b51c84cb-6f51-4c74-8c7c-057b7c7e66d3

Activity name	Activity st...	Activit...	Run start	Duration	Integration runtime	User prop...	Activity run ID
Lookup1	Succeeded	Lookup	6/27/2026, 2:45:05 PM	31s	selfhostedintegrationRuntime1		155e8ed6-4e3d-4af4-98bf-d76f2d5e040b

This is the output that we read all the table names

Output

 Copy to clipboard

```
{
  "count": 10,
  "value": [
    {
      "TABLE_NAME": "ksr_attendance"
    },
    {
      "TABLE_NAME": "ksr_batches"
    },
    {
      "TABLE_NAME": "ksr_courses"
    },
    {
      "TABLE_NAME": "ksr_enrollments"
    },
    {
      "TABLE_NAME": "ksr_exam_results"
    },
    {
      "TABLE_NAME": "ksr_exams"
    },
    {
      "TABLE_NAME": "ksr_feedback"
    },
    {
      "TABLE_NAME": "ksr_payments"
    },
    {
      "TABLE_NAME": "ksr_students"
    },
    {
      "TABLE_NAME": "ksr_trainers"
    }
  ]
}
```

The output is in key value pair

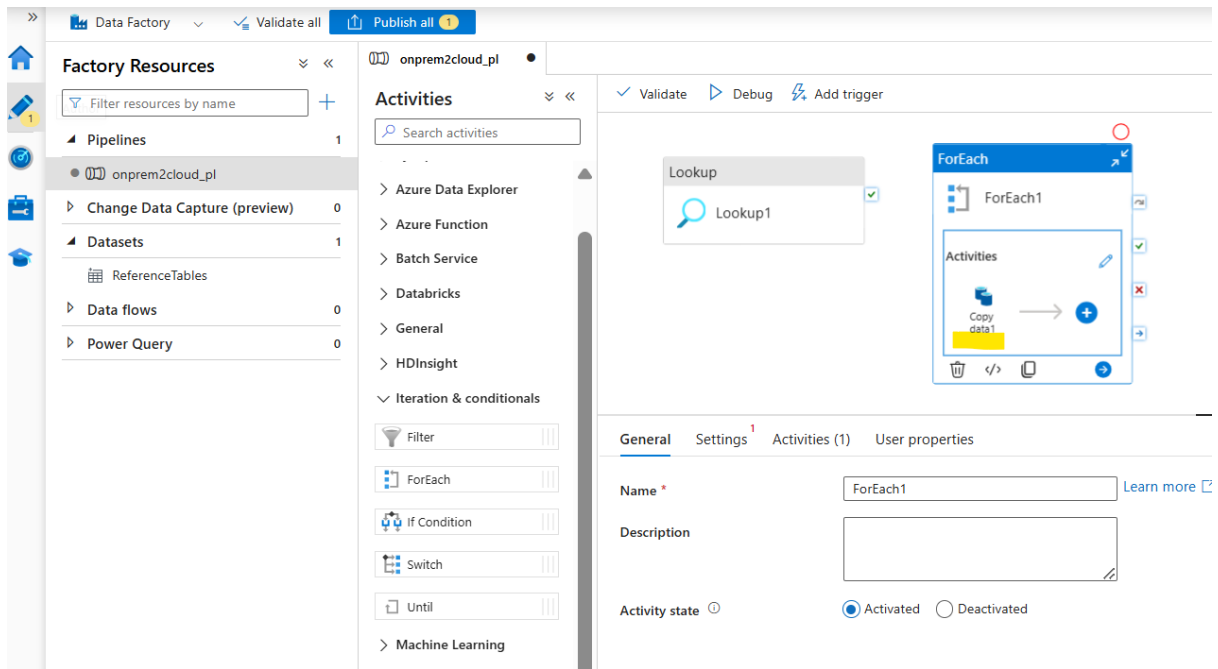
***Now we need to copy the data ----but we need to copy multiple tables so in that case---
general---iteration---for each---inside for each---1 copy activity(which runs multiple times)

The image displays two screenshots of the Microsoft Azure Data Factory (ADF) interface, specifically the 'onprem2cloud_pl' pipeline.

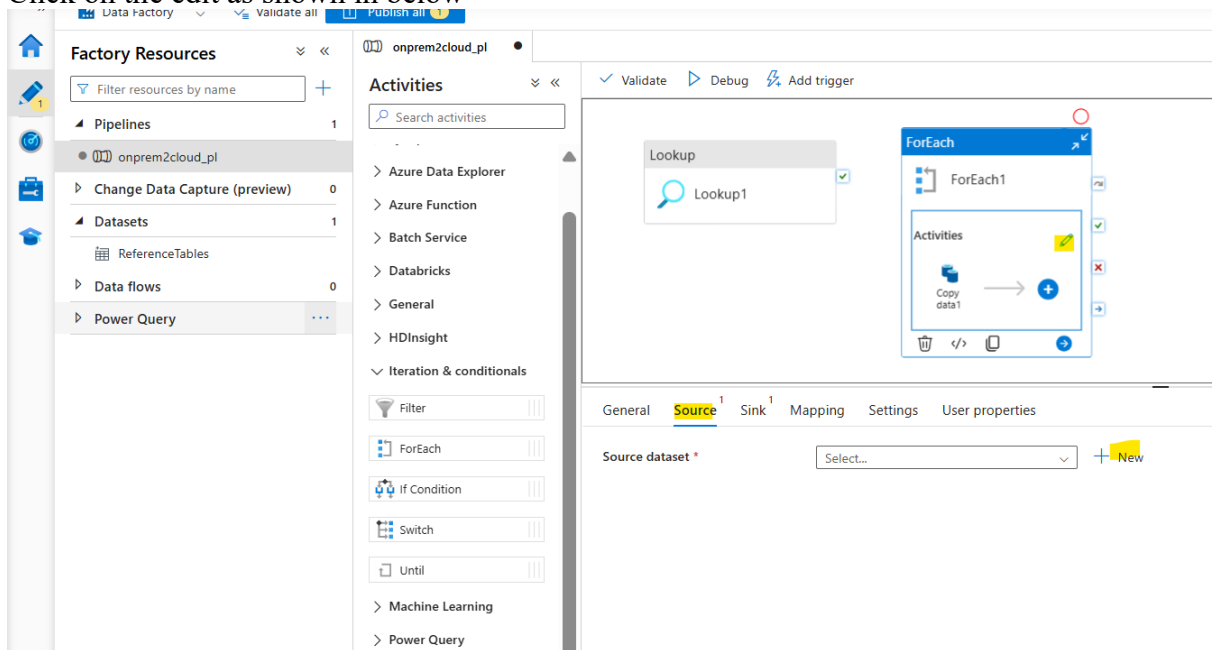
Top Screenshot: The 'Lookup' activity is selected. The 'Activities' pane on the left shows 'Lookup' under the 'Iteration & conditionals' category. The 'General' tab of the activity configuration shows the Name as 'Lookup1', Description as an empty field, Activity state as 'Activated', Timeout as '0.12:00:00', and Retry as '0'.

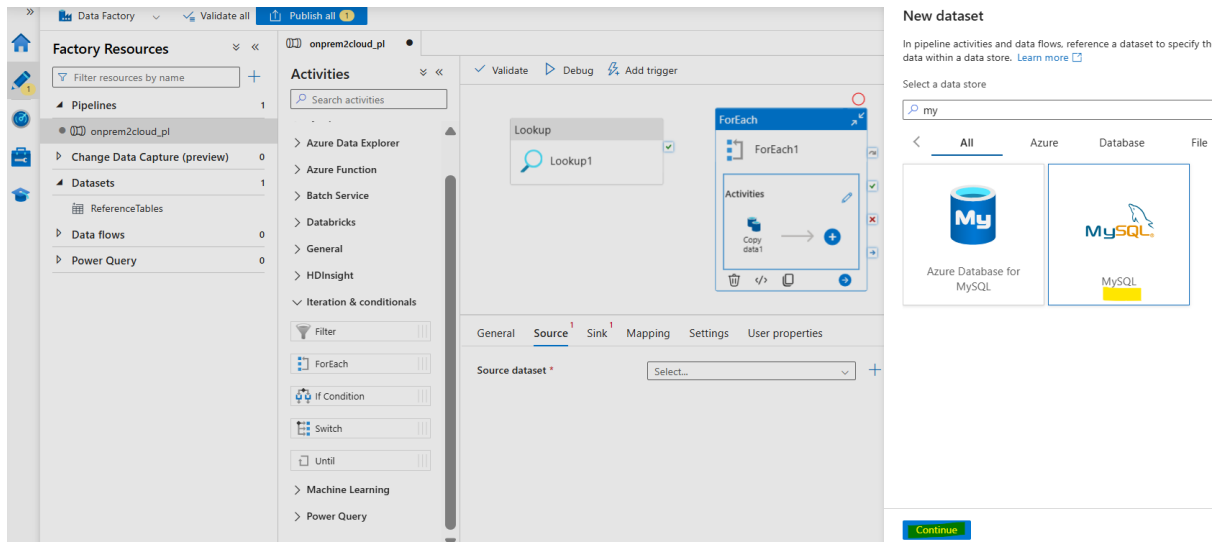
Bottom Screenshot: The 'ForEach' activity is selected. The 'Activities' pane on the left shows 'ForEach' under the 'Iteration & conditionals' category. The 'General' tab of the activity configuration shows the Name as 'ForEach1', Description as an empty field, and Activity state as 'Activated'. The 'Activities' box within the 'ForEach' activity is highlighted, showing a '+' icon, indicating where to click to add new activities.

Click + icon inside foreach to add copy data



Click on the edit as shown in below





Create a inputdata---linkedservice---don't select any data
Set properties

Name

Linked service *

Connect via integration runtime * ⓘ
☒ selfhostedintegrationRuntime1

Table name

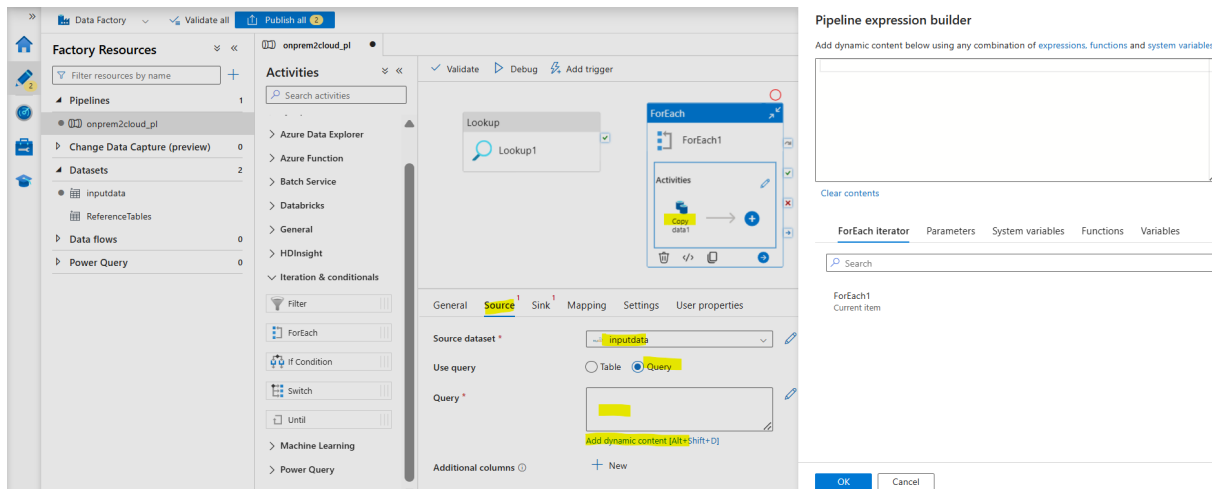
☐ Enter manually

> Advanced

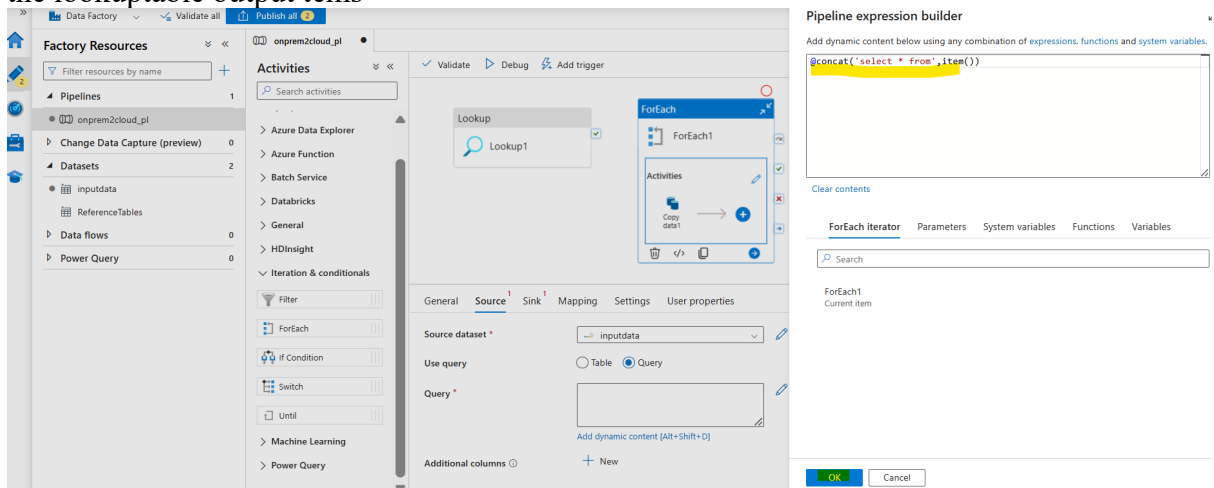
OK

Back

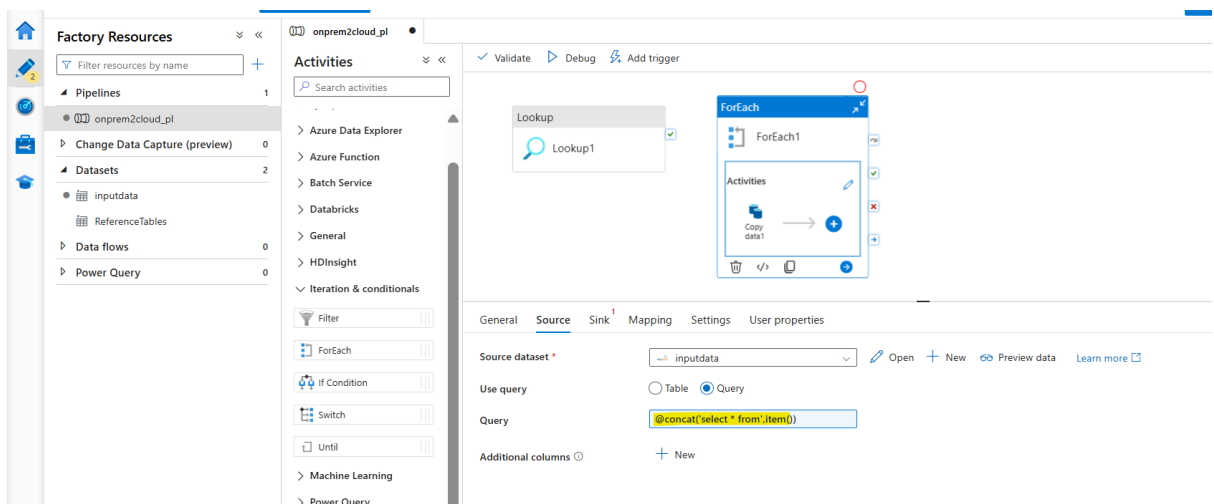
Cancel



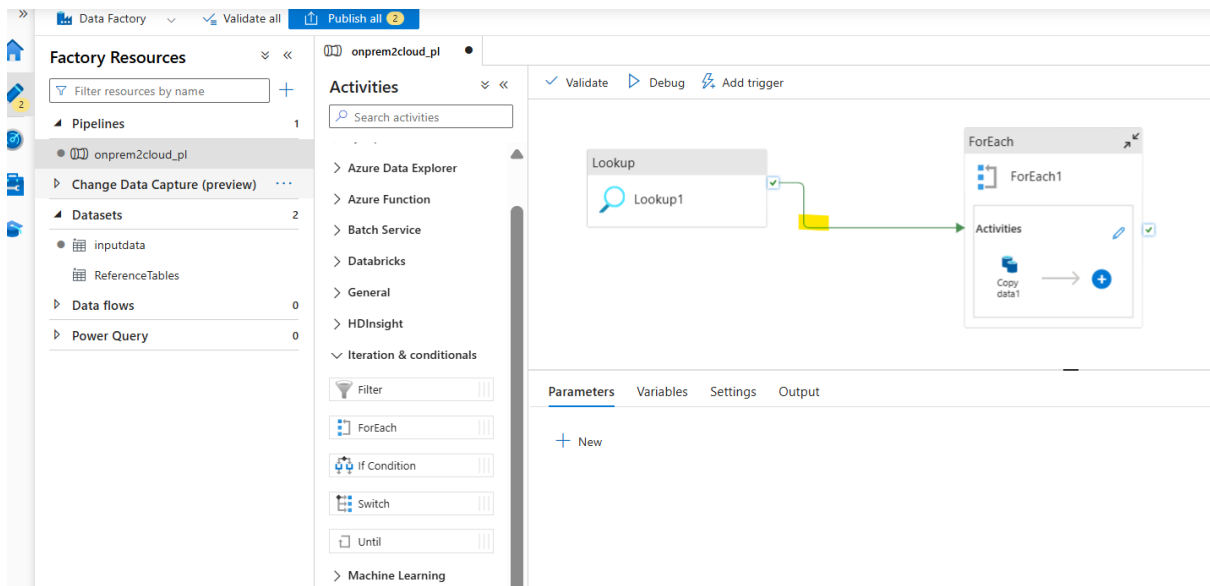
Place cursor inside the query----to show the dynamic expression mode
 Now we need to run the copy data for multiple time to read all the tbales from the database
 hence the query select * from will be same for every table and we need to add(concat)with
 the lookuptable output tems



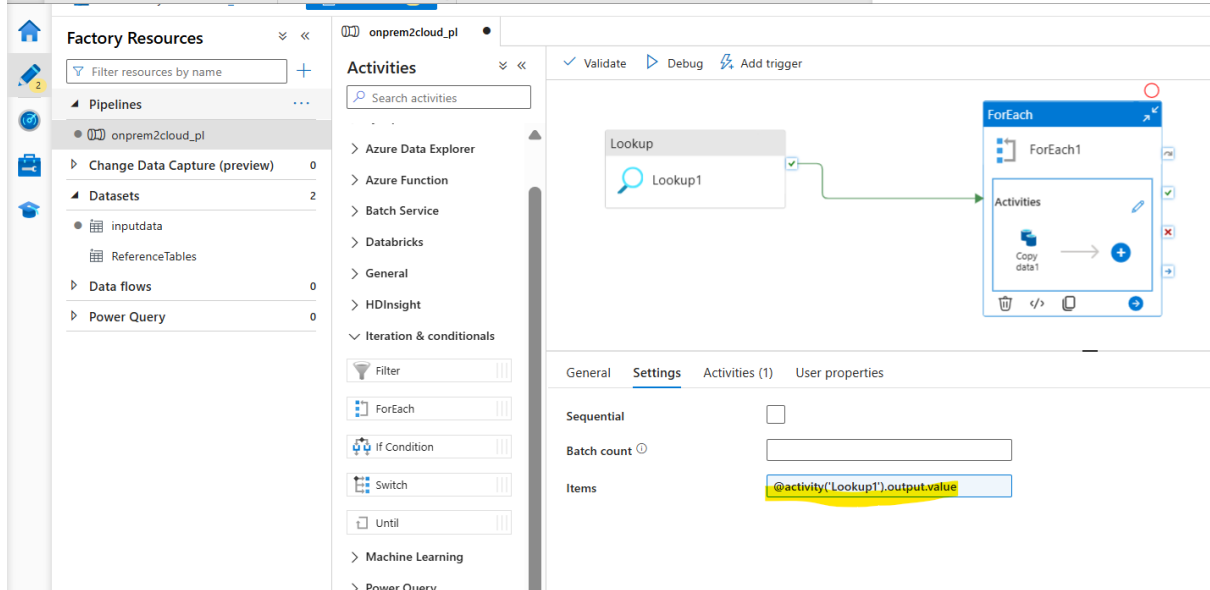
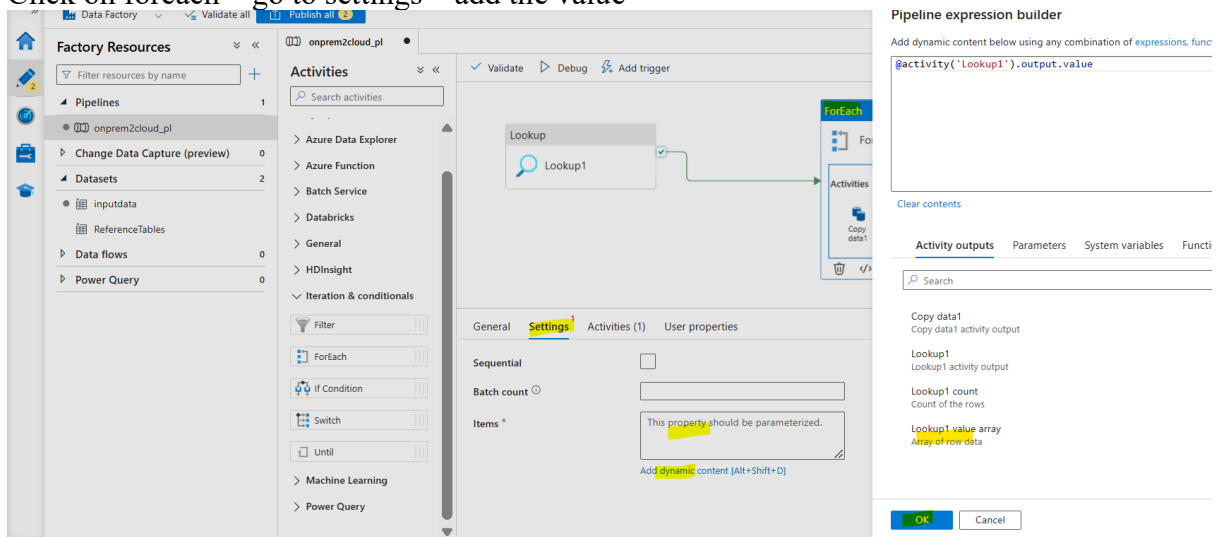
@concat('select * from',item())



Connect the lookup(on success) ----with foreach



Click on foreach---go to settings---add the value

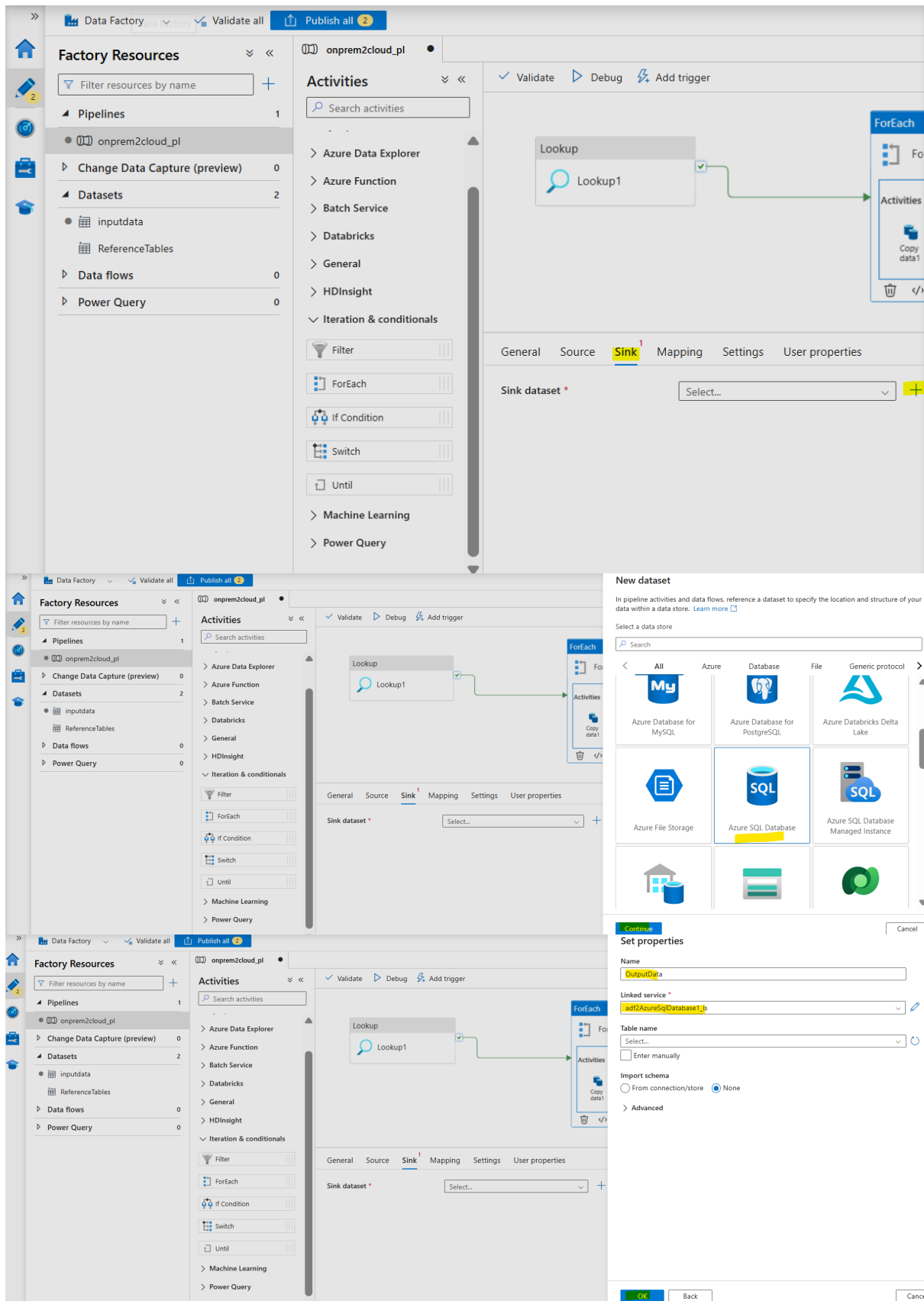


The image displays two screenshots of the Azure Data Factory (ADF) interface, showing the configuration of a pipeline activity.

Top Screenshot: The 'Pipeline expression builder' dialog is open. The 'Query' field contains the expression: `@concat('select * from', item(), 'TABLE_NAME')`. The 'Source dataset' is set to 'inputdata'. The 'Use query' radio button is selected. The 'Additional columns' section is empty.

Bottom Screenshot: The 'Lookup' activity configuration is shown. The 'Query' field contains the expression: `@concat('select * from', item(), 'TABLE_NAME')`. The 'Source dataset' is set to 'inputdata'. The 'Use query' radio button is selected. The 'Additional columns' section is empty.

We need to create a target



Let's create parameter for outputtable(or target)

» Data Factory Validate all Publish all 3

Factory Resources

Filter resources by name

- Pipelines 1
 - onprem2cloud_pl
- Change Data Capture (preview) 0
- Datasets 3
 - inputdata
 - OutputData**
 - ReferenceTables
- Data flows 0
- Power Query 0

onprem2cloud_pl • OutputData •

Azure SQL Database
OutputData

Connection Schema **Parameters**

+ New | Delete

Name	Type	Default value
outputdatatable	String	Value

» Data Factory Validate all Publish all 3

Factory Resources

Filter resources by name

- Pipelines 1
 - onprem2cloud_pl
- Change Data Capture (preview) 0
- Datasets 3
 - inputdata
 - OutputData
 - ReferenceTables
- Data flows 0
- Power Query 0

onprem2cloud_pl • OutputData •

Azure SQL Database
OutputData

Connection Schema Parameters

Linked service * adf2AzureSqlDatabase1_Is Test connection Edit + New Learn more

Table schema name @dataset().outputdatatable Preview data

☒ Enter manually

Factory Resources

Filter resources by name

- Pipelines 1
 - onprem2cloud_pl
- Change Data Capture (preview) 0
- Datasets 3
 - inputdata
 - OutputData
 - ReferenceTables
- Data flows 0
- Power Query 0

onprem2cloud_pl • OutputData •

Activities

Search activities

- Azure Data Explorer
- Azure Function
- Batch Service
- Databricks
- General
- HDInsight
- Iteration & conditionals
 - Filter
 - ForEach
 - If Condition
 - Switch
 - Until
- Machine Learning
- Power Query

Lookup1

ForEach

Copy data1

General Source **Sink** Mapping Settings User properties

Sink dataset * OutputData

Dataset properties

Name	Value
Outputdatatable	Value

Write behavior ☒ Insert ☐ Upsert ☐ Stored procedure

Bulk insert table lock ☐ Yes ☒ No

Add dynamic content below using any combination of expressions, functions and system variables

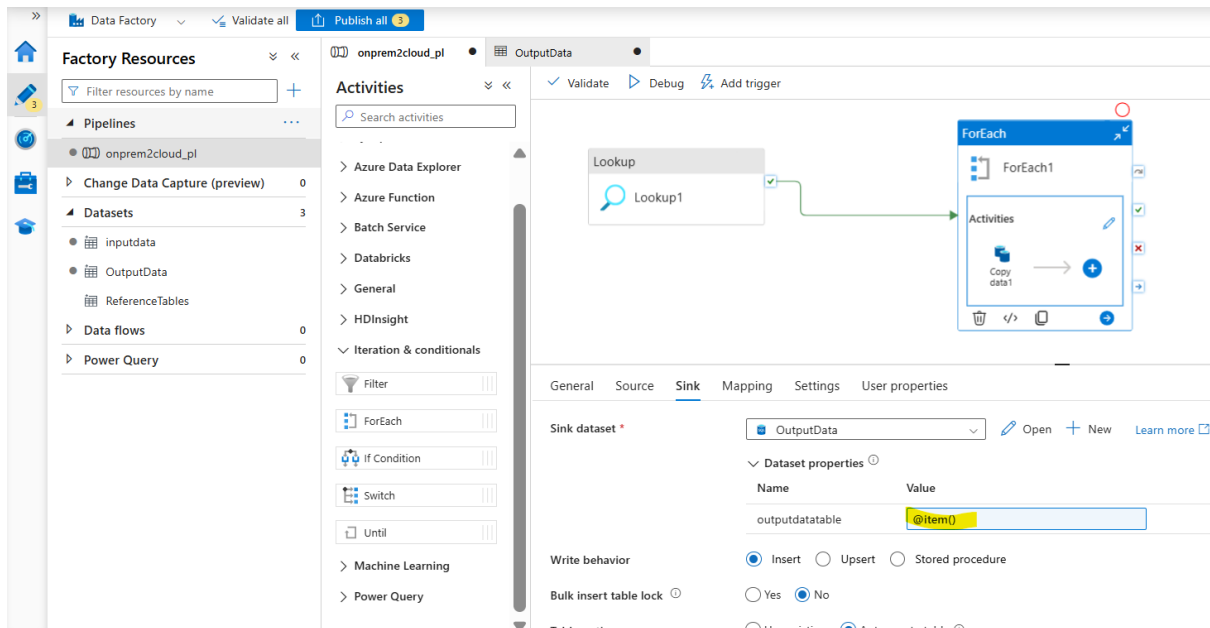
Clear contents

ForEach iterator Activity outputs Parameters System variables

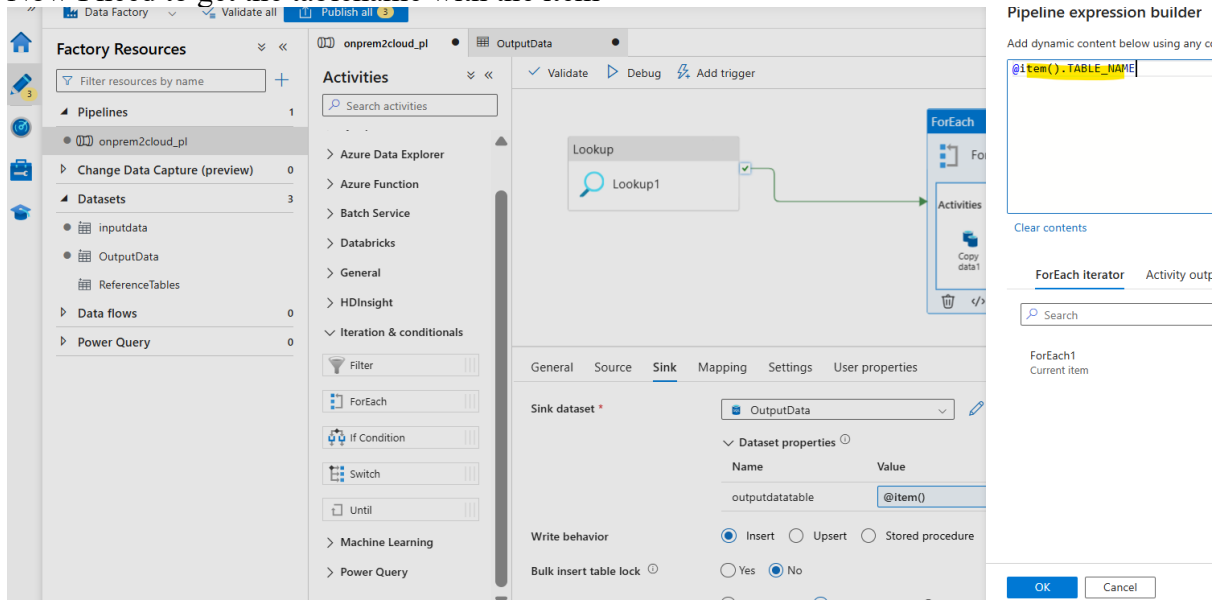
Search

ForEach1 Current item

OK Cancel



Now I need to get the tablename with the item



Factory Resources

Filter resources by name

Pipelines

- onprem2cloud_pl

Datasets

- inputdata
- OutputData
- ReferenceTables

Data flows

Power Query

Activities

Search activities

- Azure Data Explorer
- Azure Function
- Batch Service
- Databricks
- General
- HDInsight
- Iteration & conditionals
- Machine Learning
- Power Query

Filter

ForEach

If Condition

Switch

Until

Lookup

Lookup1

ForEach

ForEach1

Activities

Copy data1

General

Source

Sink

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User properties

Sink dataset *

OutputData

Open

New

Learn r

Dataset properties

Name

Value

outputdatatable

@item().TABLE_NAME

Write behavior

Insert

Upsert

Stored procedure

Bulk insert table lock

Yes

No

Table option

Use existing

Auto create table

Successfully migrated the data

Activity runs

Pipeline run ID: 65f6f6e-1146-4f61-9885-84332d0c1d14

All status

Showing 1 - 12 items

Activity name	Activity status	Activity type	Run start	Duration	Integration runtime	User properties	Activity run ID	Log
Lookup1	Succeeded	Lookup	6/27/2026, 3:35:44 PM	24s	selfhostedintegrationRuntime1		020390ba-ec72-4c2b-ad32-6a2ba8c0b7a4	
ForEach1	Succeeded	ForEach	6/27/2026, 3:36:09 PM	58s	selfhostedintegrationRuntime1		23b76eaf-5a3d-4d45-89a8-60722db5f9b0	
Copy data1	Succeeded	Copy data	6/27/2026, 3:36:10 PM	55s	selfhostedintegrationRuntime1		9330e1f-ea15-4197-8a4e-70916e63e818	
Copy data1	Succeeded	Copy data	6/27/2026, 3:36:10 PM	45s	selfhostedintegrationRuntime1		e1044bc-5c9b-478a-a9e9-c08e9949a76	
Copy data1	Succeeded	Copy data	6/27/2026, 3:36:10 PM	42s	selfhostedintegrationRuntime1		09c79b09-a15a-46a3-972b-5367754a607	
Copy data1	Succeeded	Copy data	6/27/2026, 3:36:10 PM	41s	selfhostedintegrationRuntime1		7c844fa2-182c-4640-9731-f0f0384c592	
Copy data1	Succeeded	Copy data	6/27/2026, 3:36:10 PM	41s	selfhostedintegrationRuntime1		91e5ec9f-3e78-4781-bbaf-edbd4d9e3670	
Copy data1	Succeeded	Copy data	6/27/2026, 3:36:10 PM	43s	selfhostedintegrationRuntime1		1b2f2182-809e-4937-a284-b04565a9f18a	
Copy data1	Succeeded	Copy data	6/27/2026, 3:36:10 PM	39s	selfhostedintegrationRuntime1		0b020774-c053-4288-b650-01651b30c574	
Copy data1	Succeeded	Copy data	6/27/2026, 3:36:10 PM	47s	selfhostedintegrationRuntime1		3224ec49-aec3-4160-050a-2aef1e95ec7	
Copy data1	Succeeded	Copy data	6/27/2026, 3:36:10 PM	39s	selfhostedintegrationRuntime1		90c50275-74c4-4a08-a235-490ec7a081f5	
Copy data1	Succeeded	Copy data	6/27/2026, 3:36:10 PM	46s	selfhostedintegrationRuntime1		f522cc-d5a9-4092-8412-9ca94604139	

Home > Microsoft.SQLDatabase.newDatabaseNewServer_eac6f1c5aec642308e7ee | Overview > pipelinedb (pipeline5sa/pipelinedb)

pipelinedb (pipeline5sa/pipelinedb) | Query editor (preview)

Search

New query

Templates

Open in

Feedback

Sign out

Classic experience

Overview

Activity log

Tags

Diagnose and solve problems

Query editor (preview)

Mirror database in Fabric (preview)

Resource visualizer

Settings

Data management

Integrations

Power Platform

Security

Intelligent performance

Monitoring

Automation

Help

Explorer

Search

pipelinedb

dbo

Tables

- ksr_attendance
- ksr_batches
- ksr_courses
- ksr_enrollments
- ksr_exam_results
- ksr_exams
- ksr_feedback
- ksr_payments
- ksr_students

ksr_attendance

SQL query 9

Run

Save as view

```
1 SELECT TOP (1000) * FROM [dbo].[ksr_attendance]
```

If we want to change the table names----go to sink---outputexpression(builder)---add as required

The image consists of two screenshots from the Azure Data Factory (ADF) portal, illustrating the configuration of a sink dataset to use a dynamic table name.

Top Screenshot: Shows the 'Sink' tab of the 'OutputData' dataset configuration. The 'Sink dataset' is set to 'OutputData'. The 'Write behavior' is 'Insert', and 'Bulk insert table lock' is 'No'. The 'Table option' is 'Auto create table'. The 'Sink dataset' dropdown is open, showing 'OutputData' as the selected dataset. The 'Dataset properties' section shows the 'Name' as 'outputdatatable'. The 'Pipeline expression builder' is open on the right, showing the expression: `@concat(item().TABLE_NAME, '_m_tb')`.

Bottom Screenshot: Shows the 'Sink' tab of the 'OutputData' dataset configuration. The 'Sink dataset' is set to 'OutputData'. The 'Write behavior' is 'Insert', and 'Bulk insert table lock' is 'No'. The 'Table option' is 'Auto create table'. The 'Dataset properties' section shows the 'Name' as 'outputdatatable' and the 'Value' as `@concat(item().TABLE_NAME, '_m_tb')`. The 'Pipeline expression builder' is open on the right, showing the expression: `@concat(item().TABLE_NAME, '_m_tb')`. The 'ForEach' iterator is also visible in the background.

Publish all-addtrigger---triggernow---runpipeline